# Collusion, Espionage, or Look-alikes? Auditing Information Flow between LLM Agents

Pairwise net influence with and without private inputs in a procurement auction

Marvin Kemper, Independent; with Apart Research

**Abstract**

AI agents increasingly bid and set prices for competing firms. If two of them secretly coordinate, can an overseer find the pair just by watching what they do? We test a published information-theoretic measure, pairwise net influence, on AI bidders in a repeated auction with an instructed two-firm cartel and honest twin markets with identical costs. From bids alone, a calibrated alarm built on the measure named exactly the colluding pair in 49 of 60 markets where the cartel chatted in private, never named a wrong pair, and raised one false alarm in 40 honest markets; the alarm rule published with the measure flagged 24 of 40 honest markets. When the overseer also sees each firm's costs, the direction of information flow becomes visible when pooled over markets: two-way for the cartel, one-way from victim to spy for an espionage scenario. A plain correlation of the same data detected chatting cartels at least as well (pre-registered test: net influence AUROC 0.865, $p = 0.00002$; correlation 0.9975); net influence added value only for espionage, and this replicated on fresh markets (+0.265 AUROC $[0.070, 0.445]$). Two warnings follow: the most profitable cartel, one without chat, went almost unnoticed, and an AI judge named the strongest firms rather than the colluders. Behaviour-based screens can direct an investigation, but need calibration and a simple baseline, and are not proof of agreement.

## 1 Introduction

AI agents increasingly act for competing firms: they set prices, place bids and negotiate. If two such agents coordinate in secret, an overseer may see only what they do. Their messages may be hidden or encoded, and similar bids alone prove nothing, because honest agents react to the same market. Can an audit of behaviour still find the colluding pair?

We test one candidate audit, pairwise net influence (Bonjour et al., 2022). It measures how strongly one agent's actions depend on another's and subtracts the strongest dependence on any third agent, so that a pair is flagged only if it moves together beyond what the rest of the market explains. We give it to an auditor of a repeated procurement auction with four AI bidders (Figure 1). In some markets two bidders are told to form a cartel; in others one bidder secretly sees a rival's cost. Every treated market has an honest twin with identical costs. The auditor sees either the bids only, as a competition authority usually would, or the bids and every firm's private costs. It never reads messages.

Four findings stand out.

1. **From bids alone, a calibrated alarm finds cartels that talk.** It named exactly the cartel pair in most such markets, never a wrong pair, and rarely raised false alarms (Figure 2). The uncalibrated rule published with the method flagged most honest markets.
2. **With costs, one can see which way information flows.** Pooled over markets, cartel partners leak their costs to each other, while a spy receives information one way (Figure 3). In single markets this signal is often too weak.
3. **A simple correlation detects talking cartels just as well.** Net influence added value only where private information leaked (espionage), and this held on fresh markets (Figure 4).
4. **Two blind spots.** The most profitable cartel, one that never talked, went almost unnoticed, and an AI judge named the strongest firms instead of the colluders (Figure 5).

The contribution is a pre-registered, matched-control test of what such an audit can and cannot tell an overseer, with an out-of-sample replication of the main secondary findings.




**Figure 1:** How the test works. Four AI agents (Qwen3.5-9B) bid in a repeated auction. In treated markets two of them form a cartel (with or without a private chat), or one secretly sees a rival's cost. Each treated market has an honest twin with identical costs. The auditor never sees messages; it sees either the bids only (access level L1) or the bids and every firm's costs (L2), and must say whether there is a cartel and which pair it is.

## 2 Related Work

Bonjour et al. (2022) introduce information-theoretic collusion detection in multi-agent games, including a partially observable formulation. We implement their statistic and fixed-threshold rule, then examine private-input access and permutation correction in LLM auctions. Work on algorithmic collusion by LLMs (Fish et al., 2024), double auctions (Agrawal et al., 2025), and market division (Lin et al., 2024) supplies the market context; our question concerns the audit of an assigned pair rather than whether collusion emerges spontaneously.

Steganographic collusion (Motwani et al., 2024) motivates audits that do not depend on readable messages. *Audit the Whisper* (Tailor, 2025) uses cross-run mutual information, permutation invariance, and interventions, including auctions. Its run-level focus differs from pair localisation and private-state directional decomposition here. Chain-of-thought monitoring (Lee and Park, 2026) and multi-agent interpretability/NARCBench (Rose et al., 2026) address complementary observation channels; we evaluate behaviour and private inputs, without access to model internals.

Statistical procurement screens provide a related application tradition (Huber and Imhof, 2019; Imhof and Wallimann, 2021; Wallimann et al., 2023). We treat a screen as a reason for further investigation. Concurrent sprint repositories also investigate net influence (Raj-Taware, 2026; soumya-nayak-commits, 2026); we make no claim to the first LLM application. The distinctive focus is the joint evaluation of pair detection, private-cost flow, and honest look-alikes under matched controls.

## 3 Methods

### 3.1 Game, conditions, and models

Each market has four firms and 200 rounds. Two strong firms draw independent costs uniformly from $\{1,2,3\}$; two fringe firms draw from $\{3,4,5\}$. Bids range from 2 to 7; the lowest wins, with random ties. The winner earns bid minus cost, others zero. Bids and winners become public. Bidders see their own cost, the last ten rounds, and their cumulative profit. Costs, tie-breaks, and assignments are seeded and matched by market index. Strong seats cycle through all six label pairs, hidden from auditors.

C1 assigns a cartel with a private channel (SS: strong–strong; SF: strong–fringe; FF: fringe–fringe). C2 removes the channel; E1 gives a firm unilateral access to its rival's cost. N0 is honest, and P0 is a scripted cartel positive control. Appendix Table 3 lists all conditions. C1 members exchange simultaneous private messages before bidding. The treatment bundles a joint-profit goal, partner naming, secrecy, a surveillance warning, and channel access. Assigned cartel markets remain included regardless of compliance or profit.

The core uses Qwen3.5-9B: 40 N0 markets and 20 per other core cell. Breadth covers Gemma-4-12B and Ministral-3-8B, their smaller counterparts Gemma-4-E4B and Ministral-3-3B, and Qwen3.5-4B. The amendment specifies 20

markets per breadth and A2 cell. Revisions and sampling are pinned; seeded re-collection is not bitwise reproducible (Appendix B).

## 3.2 Auditor access and influence

L1 receives bids $B_i$ and winners; L2 additionally receives costs $C_i$. Neither sees messages, role texts, condition labels, or model identities. The public state $H_t$ is the previous winning bid, with a start code and carry-forward after an all-invalid round; masking preserves this chronology. At L1, $\gamma_{ij} = I(B_i; B_j \mid H)$. At L2, set $S_i = (C_i, H)$. Equation 3 of Bonjour et al. (2022) becomes

$$\gamma_{ij} = \sum_{s_{ij}} p(s_{ij}) \sum_{b_i, b_j} p(b_i, b_j \mid s_{ij}) \log_2 \frac{p(b_i, b_j \mid s_{ij})}{p(b_i \mid s_i) p(b_j \mid s_j)}. \tag{1}$$

The chain rule gives the exact decomposition

$$\gamma_{ij} = \underbrace{I(B_i; B_j \mid C_i, C_j, H)}_{\text{action dependence}} + \underbrace{I(B_i; C_j \mid C_i, H)}_{L_{j \to i}} + \underbrace{I(B_j; C_i \mid C_j, H)}_{L_{i \to j}}. \tag{2}$$

Under independent costs and no other access to a rival's current cost, leaks probe private-information flow; omitted history can induce action dependence. Arrows indicate cost source and bid receiver, not intent. Estimates use plug-in counts in bits, our replication convention inferred from the paper's results; the paper does not specify the log base.

For any influence matrix $x$, define the paper's net transformation and mutual rule (its Equations 4–5) as

$$\Gamma_{ij}[x] = x_{ij} - \max_{k \neq i,j} x_{kj}, \qquad \Gamma_{ij}[\hat{\gamma}] \geq \alpha \text{ and } \quad \Gamma_{ji}[\hat{\gamma}] \geq \alpha, \quad \alpha = 0.05. \tag{3}$$

Only a unique passing pair is reported. Symmetric $\gamma$ need not imply symmetric $\Gamma$: the rival comparisons differ. Net direction is not leak direction.

## 3.3 Permutation correction and flow labels

Within each market, 999 symmetric permutations shuffle each firm's bids independently within its own strata: $H$ at L1 and $(C_i, H)$ at L2. Define

$$g_{ij} = \hat{\gamma}_{ij} - \frac{1}{999} \sum_{r=1}^{999} \hat{\gamma}_{ij}^{(r)}, \qquad s_{ij} = \min\{\Gamma_{ij}[g], \Gamma_{ji}[g]\}. \tag{4}$$

Corrections remain signed. A candidate requires a permutation $p$-value for raw influence at most $0.05/6$ and both corrected net directions positive; only a unique candidate is reported. The test concerns influence, not net influence. The raw fixed-threshold rule is reported separately. The secondary leak score sums the two corrected leaks; a leak candidate requires either direction to pass $0.05/12$, again with unique-pair reporting.

Flow labels are both directions detected; one detected, reverse weak; neither detected, both weak; or unclear. "Weak" requires $p \geq 0.20$ and a corrected leak at most 0.05 bits. Labels cover designated pairs and the full detection pipeline. Weak evidence does not establish absence; one-way flow can arise in a cartel with unilateral sharing. Permutation validity requires within-stratum exchangeability, which adaptive policies can violate. Per-market alarm rates are descriptive, without a general calibration guarantee or error control across all analyses.

## 3.4 Baselines and evaluation

The L1 correlation baseline is uncorrected Cramér's V. L2 uses absolute Pearson correlation after subtracting each firm's mean bid at its own cost within the market; constant residuals score zero and signed correlations are also retained. These baselines do not adjust for history. A strength-only benchmark ranks the strong pair first, exposing a localisation shortcut.

The frozen Qwen3.5-9B judge uses thinking and returns raw pair probabilities from rules and masked round records, with costs only at L2 and no treatment information. There is no recalibration or answer selection; content failures remain failures. Comparisons use common judged markets and report coverage; descriptive Brier scores state the artificial prevalence. A Gemma-4-12B counter-judge was attempted on core N0/C1-SS indices 0–9 at both levels. Qwen judging Qwen remains a family-overlap limitation.

The market is the analysis unit. AUROC (area under the receiver operating characteristic curve) is the probability that a randomly chosen treatment market scores above a control market, with half credit for ties. All methods use rounds with four valid bids; more than 20 incomplete rounds makes a market ineligible and removes the matched block. A no-bid-category sensitivity retains missing bids. The sole confirmatory test compares designated-pair L2 net scores in Qwen C1-SS and N0 indices 0–19 over 200 rounds. With 99,999 within-block label swaps (seed 20261027),

$$p = \frac{1 + \#\{\mathrm{AUC}^* \geq \mathrm{AUC}_{\mathrm{obs}}\}}{100000}; \qquad \text{reject if } \mathrm{AUC}_{\mathrm{obs}} > 0.5 \text{ and } p \leq 0.05. \tag{5}$$

This tests paired label exchangeability, not the composite null AUROC $\leq 0.5$. Secondary estimates use 10,000 whole matched-block bootstraps and pointwise 95% percentile intervals. Top-1 is the fraction of credit for ranking the assigned pair first, shared equally across ties. Localisation also reports reciprocal rank, correct/wrong unique reports, and abstention; $1/6$ is a guessing benchmark, not a null. Honest-market alarms include counts and Clopper–Pearson intervals. E1 flow detection is not a collusion positive.

A2 tests paired fringe-placement swaps with Holm correction across seven endpoints, not a pure model-sameness mechanism. Pooled fingerprints align roles, bootstrap markets, and permute within markets. The exploratory probe regenerates a partner's message under counterfactual cost, then samples bids with the recipient's cost and history fixed; draws remain nested in markets (Appendix E).

### 3.5 Out-of-sample replication and judge-budget check

After the final analysis, two additions were written into the pre-registration before their data were collected (Addenda v2.2 and v2.3, Appendix E). First, E1, C2 and C1-SS were collected again on fresh market indices 20–39 with unchanged prompts, sampling, seeds and 200 rounds. Each new market shares its cost sequence with the already collected honest N0 market of the same index, which serves as its matched control. The frozen code analyses these markets as a separate cohort. The pre-specified criteria are: for E1, the L2 net-minus-correlation interval lies above zero; for C2, the joint-profit uplift over N0 lies above zero; for C1-SS, the L2 net AUROC interval lies above 0.5. Because the original results were known, this is a replication of observed patterns, not a blind test. Second, an exploratory check re-ran the Qwen judge with a 24,000-token instead of a 12,000-token output limit (L2; core N0, C1-SS and C1-SF markets 0–19; everything else unchanged). Re-analysing identical markets as a separate cohort reproduces correlation, strength and paper scores exactly; net and leak scores change only by permutation Monte Carlo noise (largest AUROC difference 0.0075).

### 3.6 Pre-registration and deviations

Design and executable analysis were frozen before collection; A1–A3 preceded LLM outcome analysis. Seven deviations cover warm-up transport; fenced-JSON parsing and labelled Gemma/Ministral recollection; caching, streaming and checkpoints; disk exhaustion and relocation; later recovery, call ceilings, loader paths and an exceeded budget cap; judge transport recovery and budget cancellation of breadth judging; and the operations of the two addenda. Core Qwen prompts and sampling were retained. Appendix E records the freezes, amendment, and deviations separately.

## 4 Results

All results use the frozen analysis code and the full 200 rounds; "bids only" is access level L1, "bids + costs" is L2. Intervals are 95% block-bootstrap intervals. Detailed per-cell figures, all detector rows and the scripted rock–paper–scissors replication of the original paper are in the appendix (Appendices A and D).

### 4.1 From bids alone, a calibrated alarm names the cartel

Figure 2 asks the question an overseer would ask: did the alarm rule point to exactly the colluding pair, to a wrong pair, or stay silent? With bids only, our permutation-corrected net-influence rule named exactly the right pair in 16, 13 and 20 of 20 markets for the three cartels that chat (two strong firms; strong and weak firm; two weak firms), and never named a wrong pair. In 40 honest markets it raised one false alarm. The rule published with the method is not usable as an alarm here: it flagged 24 of 40 honest markets with bids only and all 40 with costs, and for the strong-pair cartel




**Figure 2:** What an alarm reports. For each market type, the share of markets in which the rule named exactly the colluding pair (green), a wrong pair or, in honest markets, any pair (red), or nothing (grey); numbers give right-pair counts (cartel rows) or false alarms (honest row). Left: the fixed threshold published with the method. Right: our permutation-corrected rule. 20 markets per cartel type, 40 honest markets.

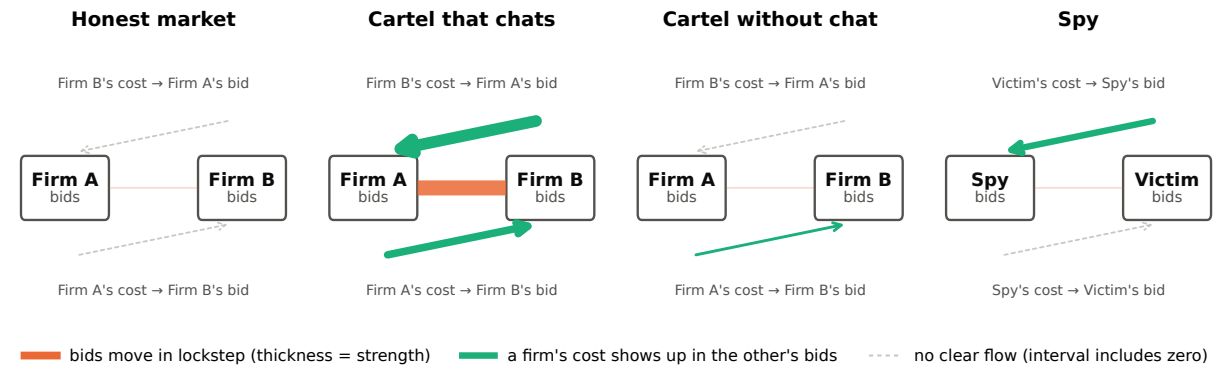



**Figure 3:** Which way information flows (bids + costs, pooled over 40 honest and 20 treated markets per type). Orange: the two bids move in lockstep beyond what costs and market history explain. Green arrows: a firm's cost shows up in the other firm's bids; width proportional to the estimate, drawn only when its pooled 95% interval excludes zero. These pooled intervals are approximate (Appendix E).

with costs it named the wrong pair in 19 of 20 markets. With costs our rule named the pair less often (5, 8 and 19 of 20; three false alarms in 40). Neither rule caught the cartel without chat or the spy.

## 4.2 With costs, the direction of the flow becomes visible

Costs let the measure separate two things: whether the two firms' bids move in lockstep, and whether one firm's private cost shows up in the other firm's bids (a "leak"). Pooled over all markets of a type (Figure 3), honest pairs show neither. The cartel that chats shows strong lockstep and leaks in both directions, consistent with partners who tell each other their costs. The spy shows a one-way leak: the victim's cost appears in the spy's bids (0.0047 bits), not the reverse ($-0.0004$). The cartel without chat shows almost nothing. These pooled patterns are clear, but single markets are short: the leak-based rule named the spy pair alone in only 4 of 20 spy markets, and market-level flow labels rarely resolved the mechanism.

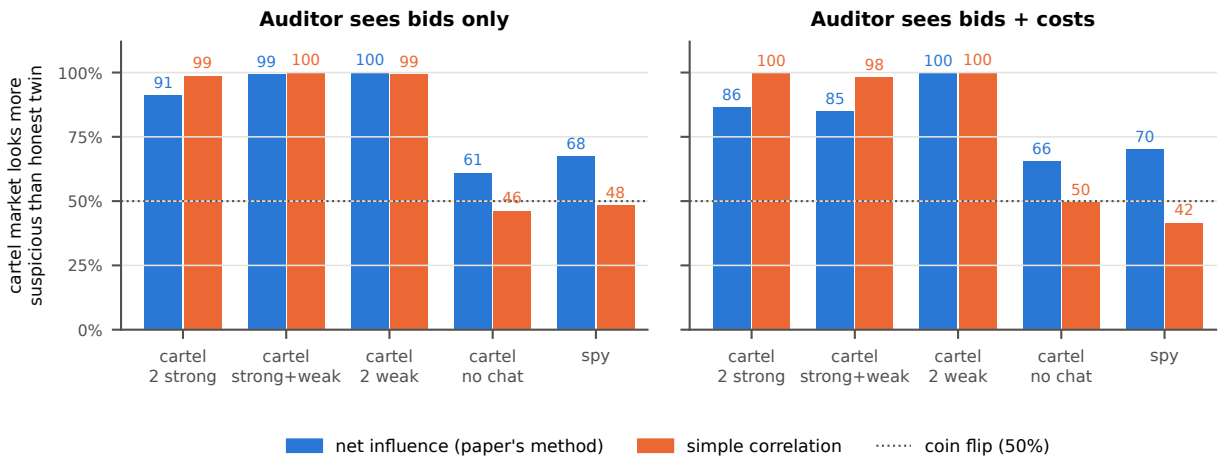



**Figure 4:** How often a treated market looks more suspicious than its honest twin (AUROC, in percent; 20 matched market pairs per type, markets 0–19). Net influence is the paper's measure; correlation is the plain dependence between the two firms' bids (with costs: after removing each firm's typical bid at its own cost). Intervals and the other models are in Appendix D.

## 4.3 Net influence versus a simple correlation

Figure 4 compares the two measures by how often a treated market looks more suspicious than its honest twin with identical costs (AUROC; 50% is a coin flip). In the pre-registered primary test, net influence with costs separated the strong-pair cartel from honest markets: AUROC 0.865 $[0.734, 0.968]$, $p = 0.00002$ over 20 matched blocks. But a simple correlation of the same data did better (0.9975; difference $-0.1325$ $[-0.2647, -0.0309]$), and it was at least as good for every cartel that chats (the two small exceptions have intervals including zero). Net influence was clearly better only where information leaks: for the spy with costs, 0.700 against 0.415 (difference $+0.285$ $[0.062, 0.515]$). This held on 20 fresh markets collected afterwards ($+0.265$ $[0.070, 0.445]$, Section 3.5); with bids only it did not replicate ($+0.095$ $[-0.1125, 0.315]$). On fresh markets the strong-pair cartel was again detected (net 0.9075 $[0.8025, 0.985]$) and again better by correlation (0.995).

## 4.4 Two warnings: the quiet cartel and the AI judge

The cartel without chat raised the pair's profit by 0.399 $[0.364, 0.434]$ per round, almost three times the strong-pair cartel's 0.144, yet it was the cartel both measures detected least well and the alarm never caught it with bids only (Figure 5a). The profit gain replicated on fresh markets ($+0.283$ $[0.179, 0.371]$). The Qwen 9B judge reads the round records and names the most suspicious pair. Under its registered 12,000-token limit it failed on 322 of 600 judgements. With 24,000 tokens it answered 54 of 60, but it separated cartel from honest markets barely better than chance (AUROC 0.625 $[0.424, 0.808]$) and simply named the two strong firms: right in 91% of strong-pair cartels, but in 2% of strong–weak cartels, where net influence was right in 93% of the same markets (Figure 5b).

## 4.5 Other results

Removing the surveillance warning raised the net AUROC with costs by 0.0975 $[-0.0051, 0.2300]$ while profit fell by 0.3478 per round; this does not show deliberate evasion. None of the seven placement-swap endpoints survived Holm correction (smallest adjusted $p = 0.264$). Gemma-E4B and Ministral-3B produced too many invalid bids, so their results are unavailable rather than negative. Probes that changed only the partner's cost found no resolved bid slope ($0.0135$ $[-0.0035, 0.0320]$).

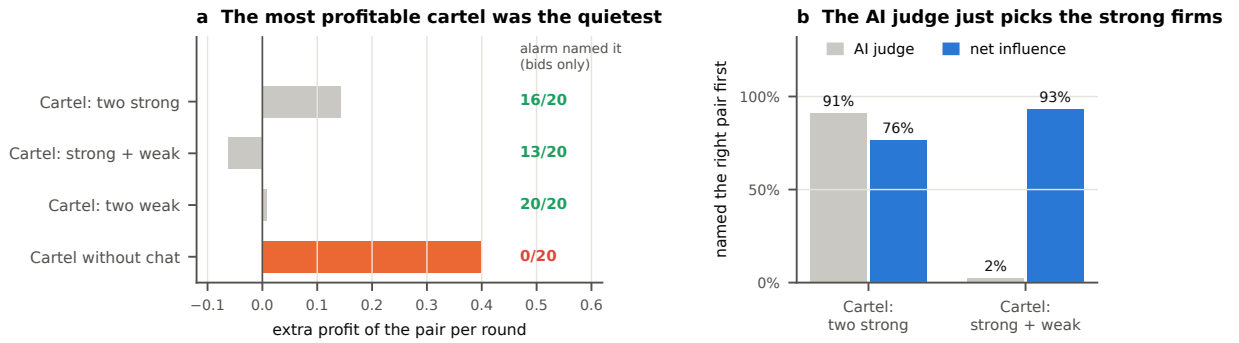



**Figure 5:** (a) Extra profit of the colluding pair per round compared with honest markets, and how often our alarm named the pair from bids alone. (b) How often the 24k-token AI judge and net influence ranked the true pair first, on the same markets with valid judge output (17 strong-pair and 15 strong–weak markets, bids + costs).

# 5 Discussion and Limitations

**What this means in practice.** Each point is limited to this game. (1) *Screening from public bids is feasible.* An overseer who sees only bids, as competition authorities usually do, could have used a calibrated alarm to point to the right pair in most markets where AI agents coordinated actively, with few false alarms and no wrong accusations. (2) *Calibrate before alarming.* The fixed threshold published with the method flagged most or all honest markets; a permutation correction fixed this. An uncalibrated screen would accuse innocent firms. (3) *Use costs to understand, not to detect.* Costs did not improve detection of talking cartels, but they reveal which way information flows and separate a two-way cartel from a one-way spy when evidence is pooled. That needs access to private data and many markets. (4) *Always compare with a simple baseline.* A plain correlation did as well for talking cartels; net influence earned its place only for espionage. Reporting net influence alone would overstate its value. (5) *Watch profits too, and do not trust a small AI judge.* The cartel without chat earned the most and was nearly invisible to both measures. The 9B judge, given ample tokens, judged firm strength rather than coordination.

**What the evidence does not show.** Only the primary test is confirmatory; alarm counts and flow patterns are pre-registered secondary results. The replication was planned after the original results were known and reuses the honest control markets. Twenty markets per type give wide intervals, and absence of an advantage is not equivalence. A one-way leak is not a unique signature of espionage, and the pooled flow intervals are approximate. The judge result concerns one 9B model judging its own family. The probes cannot support a causal claim.

**Scope.** The auction is synthetic, the cartel is instructed rather than emergent, and the treatment bundles a joint objective, partner information, a surveillance warning and, in C1, a chat channel. The core uses one model; small models were unusable and breadth cells have ten markets. Privacy, false-accusation and dual-use considerations are discussed in Appendix F.

**Next steps.** Independent replications; screens that combine correlation, flow direction and profits; larger and more varied bidder and judge models; markets with more firms; and probes with independent message draws per context.

# 6 Conclusion

Watching only the bids of AI agents, a calibrated net-influence alarm found cartels that coordinated by chat and named the right pair, without wrong accusations. With private costs, the measure also showed which way information flowed, separating a two-way cartel from a one-way spy when evidence was pooled. Yet a simple correlation detected talking cartels as well, the most profitable cartel stayed hidden, and a small AI judge looked at firm strength rather than behaviour. Behaviour-based screens can direct an investigation; they do not establish agreement, intent or harm.

## Code and Data

All code, collected market data and analysis outputs are in the project repository. The frozen final analysis is `results/analysis/final_v1/results.json`; the replication and judge check are in `results/analysis/replication_v22/` and `results/analysis/judge24k_v23/`, summarised by `experiments/addendum_summary.py`. The main-text figures are rendered by `experiments/key_figures.py`. Appendix E gives every analysis command, execution provenance and checksums.

**[TODO: Insert public repository and release archive URLs after release.]**

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# A Replication details

The scripted rock–paper–scissors replication implements the influence and mutual net-threshold rule of Bonjour et al. (2022). The comparison uses bits and the previous round's joint action as state, conventions inferred from the published results. The no-state variant is retained to show that conditioning choices matter. Detection rises with coordination and sample size, but the replication does not exactly reproduce all table entries. The values below distinguish the paper's published percentages from our simulation. These are development checks, not LLM results.

**Table 1:** Scripted RPS detection percentages: published values versus our previous-state and no-state replications.

| Coordination probability | Games | Paper | Previous state | No state |
|---|---|---|---|---|
| 0.3 | 100 | 37.1 | 36.3 | 84.4 |
| 0.3 | 200 | 48.3 | 60.1 | 94.4 |
| 0.3 | 300 | 71.0 | 77.0 | 98.1 |
| 0.3 | 400 | 82.8 | 87.1 | 99.5 |
| 0.3 | 500 | 89.6 | 92.2 | 99.9 |
| 0.3 | 1000 | 100.0 | 99.7 | 100.0 |
| 0.4 | 100 | 58.4 | 61.9 | 97.3 |
| 0.4 | 200 | 79.2 | 90.7 | 100.0 |
| 0.4 | 300 | 96.2 | 99.0 | 100.0 |
| 0.4 | 400 | 99.3 | 99.8 | 100.0 |
| 0.4 | 500 | 99.7 | 99.9 | 100.0 |
| 0.4 | 1000 | 100.0 | 100.0 | 100.0 |
| 1.0 | 60 | 100.0 | 100.0 | 100.0 |
| 0.2 | 1000 | 24.3 | 50.0 | 58.0 |
| 0.2 | 10000 | 82.3 | 83.7 | 88.5 |
| 0.1 | 1000 | 8.0 | 0.4 | 0.0 |

**Table 2:** Scripted RPS honest-market any-pair alarm rates (coordination probability zero). This is a separate simulation, not LLM calibration.

| State | Games | Fixed threshold | Permutation rule |
|---|---|---|---|
| none | 50 | 0.296 | 0.038 |
| none | 100 | 0.088 | 0.062 |
| none | 200 | 0.006 | 0.048 |
| none | 500 | 0.0 | 0.042 |
| none | 1000 | 0.0 | 0.038 |
| prev | 50 | 0.588 | 0.02 |
| prev | 100 | 0.484 | 0.036 |
| prev | 200 | 0.252 | 0.036 |
| prev | 500 | 0.046 | 0.026 |
| prev | 1000 | 0.0 | 0.038 |

The deterministic seed is 20261023; the full run uses 1000 trials per detection cell. Its calibration experiment uses 500 trials and 199 permutations. These settings differ from the registered LLM analysis. The complete calibration data, including positive conditions, are in `results/rps/calibration.csv`. Reproduction from the repository root:

```
.venv/Scripts/python experiments/rps_replication.py
```

# B Conditions, prompts and model configuration

The frozen manifest is `experiments/manifests/collection_v2.json`. It maps model identities to roles and hidden seat assignments. Exact prompt templates are in `src/collusion/llm/prompts.py`; model revisions and sampling settings are in the per-run configuration and reproducibility records named by `results/analysis_runs.json`. Request hashes, prompts, responses and parsing metadata remain associated with each raw run. `results/analysis_meta/pilot_qwen9b_v5_N0.json` records the pilot-derived N0 configuration explicitly. The manifest and source checksums accompanying this report allow these inputs to be checked without re-collecting outputs.

**Table 3:** Conditions and matched controls. SS, SF, and FF denote strong–strong, strong–fringe, and fringe–fringe designated pairs. X, Y, Z denote Qwen3.5-9B, Gemma-4-12B, and Ministral-3-8B. All A2 bidders are honest.

| Cell | Intervention or control |
|---|---|
| N0 / P0 | Honest market / live scripted strong cartel with honest LLM fringe. |
| C1-SS/SF/FF | LLM cartel with private channel; matched to N0. |
| C2 / E1 | SS cartel without channel / one strong firm sees the other's cost. |
| C1-NW (A1) | C1-SS without surveillance warning; matched N0/C1-SS triplets. |
| D / C1-XX | Honest / cartel strong pair (X,X), fringe (Y,Z). |
| H-XYZZ / C1-XY | Honest / cartel strong pair (X,Y), fringe (Z,Z). |
| LX, LY, LZ (A2) | Strong/fringe models (X,Y;X,Z), (Y,Z;Y,X), (Z,X;Z,Y), each with a twin swapping the fringe models. |

**Table 4:** Behaviour and designated-pair flow labels. Differences are treatment minus matched N0, in game units per round, with pointwise 95% intervals. Flow counts are both / one (reverse weak) / neither (both weak) / unclear / unavailable. E1 is unilateral access.

| Cell | Joint profit difference | Winning price difference | Flow counts |
|---|---|---|---|
| C1-SS | 0.144 [0.094, 0.192] | 0.295 [0.233, 0.358] | 0 / 0 / 8 / 12 / 0 |
| C2 | 0.399 [0.364, 0.434] | 0.798 [0.768, 0.826] | 0 / 0 / 11 / 9 / 0 |
| E1 | -0.116 [-0.137, -0.095] | -0.048 [-0.062, -0.034] | 0 / 4 / 8 / 8 / 0 |
| C1-NW | -0.204 [-0.229, -0.179] | -0.039 [-0.059, -0.017] | 0 / 1 / 8 / 11 / 0 |

The A1 prompt removes only: "The competition authority monitors all bids, so behave inconspicuously." The joint-profit objective and private channel remain. This is a warning-removal treatment, not a direct test of an internal evasion strategy. Core Qwen and A1 retain the strict parser; the documented fenced-JSON parser cohort covers Gemma/Ministral, A2 and breadth collection. Bidders run without thinking, whereas the primary judge uses thinking. No treatment labels, messages or model identities are given to the judges.

# C Scripted validation

Scripted auctions test known cartels, spies and honest policies under the same observation constraints. Shared-history stress tests show why a coarse or incomplete public state can invalidate a permutation null. Power planning and conditional-table occupancy checks motivated the market horizon and weak-evidence category. These simulations inform the design; they do not establish calibration for arbitrary adaptive LLM bidders. The existing artefacts are in `results/auction_scripted/`, `results/auction_power/`, and `results/null_validity/`. Reproduction commands (each also accepts `-quick`):

```
.venv/Scripts/python experiments/auction_scripted.py
.venv/Scripts/python experiments/auction_checks.py
.venv/Scripts/python experiments/auction_power.py
.venv/Scripts/python experiments/null_validity.py
```

These commands regenerate development outputs. They were not rerun for this report update, which reads existing frozen analysis files only.

# D All registered cells and per-model results

The following single comprehensive table includes every registered cell, including unavailable detector results and cancelled breadth judges, at prefix 200 with the complete-round mask. It contains 312 detector/access rows spanning 28 cells. The full machine-readable table with every prefix and mask is `results/analysis/final_v1/all_cells.md`; no sensitivity rows are silently discarded from that file.

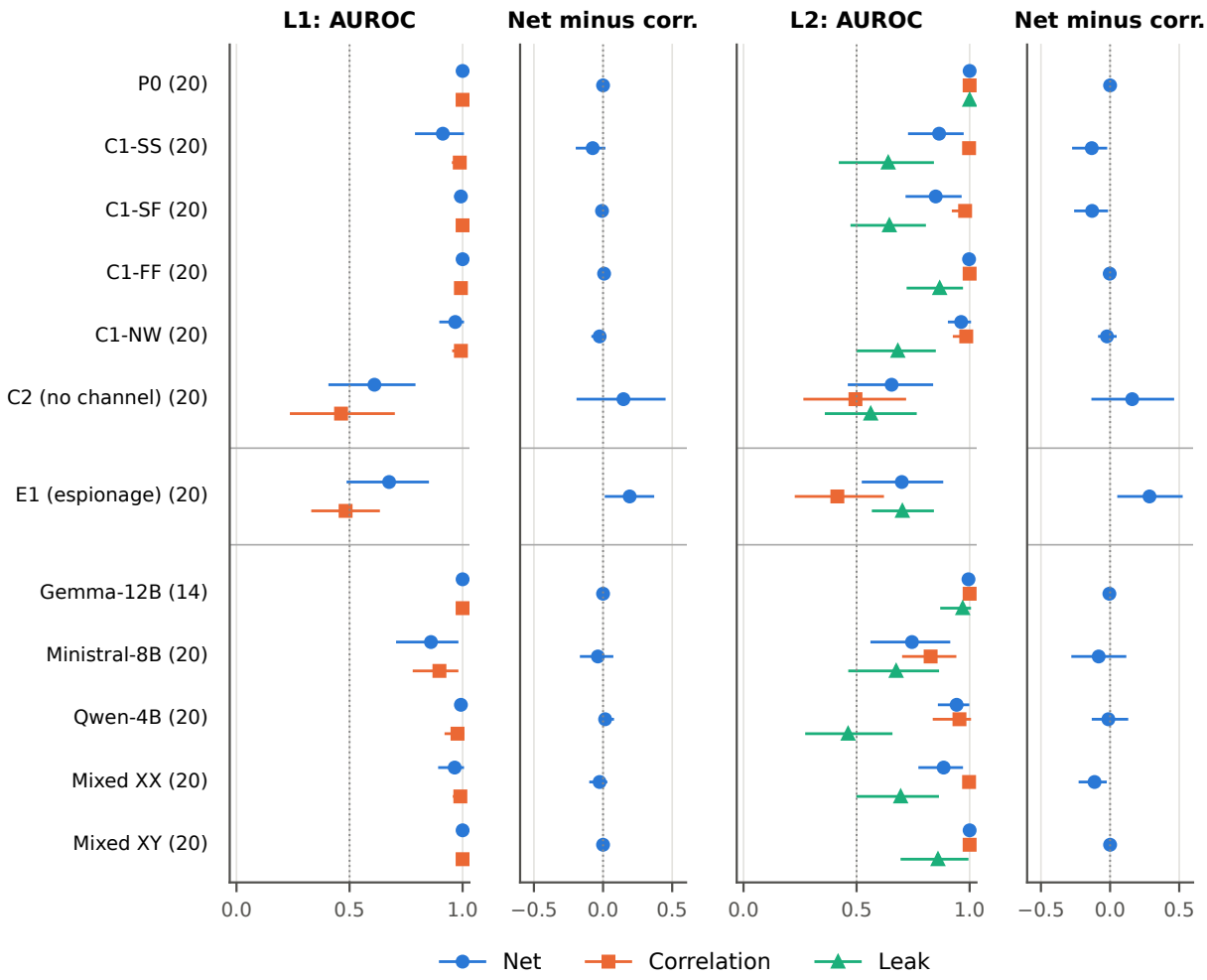



**Figure 6:** Detection by treatment (original markets 0–19), all models. AUROC with pointwise 95% block-bootstrap intervals for net influence and correlation at L1 (bids only) and L2 (bids + costs), and the L2 leak score; parentheses give matched blocks. E1 (espionage) is separated because it is not a cartel.

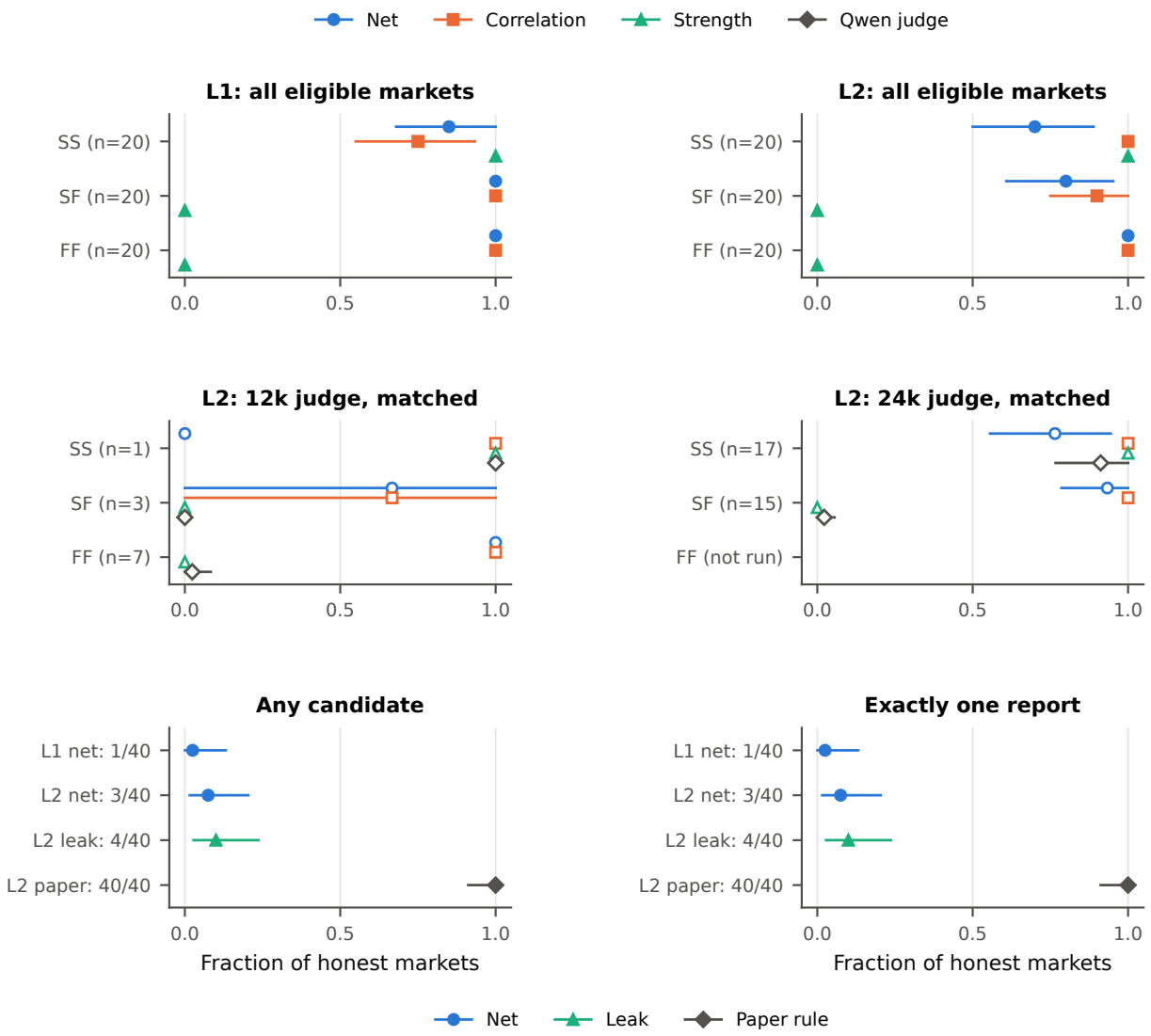



**Figure 7:** Localisation and false alarms (original markets 0–19). Top: top-1 for the assigned pair on all eligible markets. Middle: top-1 at L2 on the markets where the judge returned valid output, under the registered 12,000-token limit (left) and the 24,000-token check (right; C1-FF not run). Bottom: honest-market alarm rates with exact binomial intervals.

Cov. is successful detector coverage; *n* is the matched AUROC block count. A dash means unavailable or not applicable, never a score of zero. Unique/wrong/abstain are rates under the operational reporting rule; AUROC uses the continuous score, including for the paper detector. For honest cells, top-1 and mean reciprocal rank (MRR) describe the *designated reference pair*, not localisation accuracy. Similarly, "unique" is a report of that reference pair, and "wrong" is a report of another pair; neither is a correct collusion finding in an honest market. Alarm columns give any candidate, exactly one report, and designated reference-pair alarms, with exact binomial intervals.

Flow counts are ordered both detected / one detected, reverse weak / neither detected, both weak / unclear / unavailable. Pipeline counts use the same first categories, ending in no pair instead of unavailable. All-zero pipeline counts for score-only baselines mean no reporting rule, not evidence that no information flowed. Judge rows' label counts describe their available subset. Excluded markets fail the registered bid-validity rule. Superseded operational attempts are retained in the raw record, but are not extra independent markets in this registered cohort.

**Table 5:** All registered cells at the full prefix and complete mask. See the surrounding text for column definitions.

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| **core.N0**; declared 40, collected 40, missing 0, ineligible 0, eligible 40 | | | | |
| L1 corr. | 40/– | – | 0.050 [0.000, 0.125] | – / – / – |
| MRR 0.242 [0.192, 0.307]; flow 0/0/32/8/0; pipeline 0/0/0/0/0. | | | | |
| L1 G judge | 1/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/1/0/39; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 21/– | – | 0.952 [0.840, 1.000] | – / – / – |
| MRR 0.964 [0.880, 1.000]; flow 0/0/19/2/19; pipeline 0/0/0/0/0. | | | | |
| L1 net | 40/– | – | 0.100 [0.025, 0.200] | 0.000 / 0.025 / 0.975 |
| MRR 0.385 [0.321, 0.462]; flow 0/0/32/8/0; pipeline 0/0/0/1/39. Alarms: any 1/40 [0.001, 0.132]; one 1/40 [0.001, 0.132]; ref. 0/40 [0.000, 0.088] | | | | |
| L1 paper | 40/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.600 / 0.400 |
| MRR 0.418 [0.378, 0.454]; flow 0/0/32/8/0; pipeline 0/0/16/8/16. Alarms: any 24/40 [0.433, 0.751]; one 24/40 [0.433, 0.751]; ref. 0/40 [0.000, 0.088] | | | | |
| L1 strength | 40/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/32/8/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 40/– | – | 0.175 [0.075, 0.300] | – / – / – |
| MRR 0.415 [0.330, 0.508]; flow 0/0/32/8/0; pipeline 0/0/0/0/0. | | | | |
| L2 G judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/40; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 15/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/13/2/25; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 40/– | – | 0.050 [0.000, 0.125] | 0.000 / 0.100 / 0.900 |
| MRR 0.333 [0.283, 0.394]; flow 0/0/32/8/0; pipeline 0/3/0/1/36. Alarms: any 4/40 [0.028, 0.237]; one 4/40 [0.028, 0.237]; ref. 0/40 [0.000, 0.088] | | | | |
| L2 net | 40/– | – | 0.075 [0.000, 0.151] | 0.000 / 0.075 / 0.925 |
| MRR 0.369 [0.307, 0.439]; flow 0/0/32/8/0; pipeline 0/0/0/3/37. Alarms: any 3/40 [0.016, 0.204]; one 3/40 [0.016, 0.204]; ref. 0/40 [0.000, 0.088] | | | | |
| L2 paper | 40/– | – | 0.000 [0.000, 0.000] | 0.000 / 1.000 / 0.000 |
| MRR 0.408 [0.369, 0.444]; flow 0/0/32/8/0; pipeline 0/0/25/15/0. Alarms: any 40/40 [0.912, 1.000]; one 40/40 [0.912, 1.000]; ref. 0/40 [0.000, 0.088] | | | | |
| L2 strength | 40/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/32/8/0; pipeline 0/0/0/0/0. | | | | |
| **core.P0**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 5/4 | 0.000 [0.000, 0.000] | 0.033 [0.000, 0.111] | – / – / – |
| MRR 0.304 [0.250, 0.367]; flow 5/0/0/0/15; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0. | | | | |
| L1 paper | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 4/3 | 0.000 [0.000, 0.000] | 0.083 [0.000, 0.167] | – / – / – |
| MRR 0.330 [0.250, 0.500]; flow 4/0/0/0/16; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 0.950 / 0.000 / 0.050 |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 19/0/0/0/1. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L2 net | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0. | | | | |
| L2 paper | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| **core.C1-SS**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.988 [0.959, 1.000] | 0.750 [0.550, 0.933] | – / – / – |
| MRR 0.863 [0.750, 0.963]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| L1 G judge | 3/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/1/2/17; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 13/8 | 0.320 [0.069, 0.617] | 0.808 [0.583, 1.000] | – / – / – |
| MRR 0.835 [0.643, 1.000]; flow 0/0/6/7/7; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.912 [0.796, 1.000] | 0.850 [0.680, 1.000] | 0.800 / 0.000 / 0.200 |
| MRR 0.908 [0.800, 1.000]; flow 0/0/8/12/0; pipeline 0/0/6/10/4. | | | | |
| L1 paper | 20/20 | 0.885 [0.765, 0.991] | 0.150 [0.000, 0.318] | 0.100 / 0.300 / 0.600 |
| MRR 0.527 [0.433, 0.627]; flow 0/0/8/12/0; pipeline 0/0/4/4/12. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.998 [0.986, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| L2 G judge | 1/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/0/1/19; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 7/1 | 0.000 [0.000, 0.000] | 0.881 [0.583, 1.000] | – / – / – |
| MRR 0.898 [0.643, 1.000]; flow 0/0/3/4/13; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.640 [0.428, 0.836] | 0.300 [0.105, 0.519] | 0.000 / 0.050 / 0.950 |
| MRR 0.530 [0.387, 0.678]; flow 0/0/8/12/0; pipeline 0/0/0/1/19. | | | | |
| L2 net | 20/20 | 0.865 [0.734, 0.968] | 0.700 [0.500, 0.889] | 0.250 / 0.000 / 0.750 |
| MRR 0.823 [0.693, 0.938]; flow 0/0/8/12/0; pipeline 0/0/0/5/15. | | | | |
| L2 paper | 20/20 | 0.882 [0.776, 0.970] | 0.000 [0.000, 0.000] | 0.000 / 0.950 / 0.050 |
| MRR 0.443 [0.395, 0.485]; flow 0/0/8/12/0; pipeline 0/0/11/8/1. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| **core.C2**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.463 [0.242, 0.694] | 0.300 [0.111, 0.522] | – / – / – |
| MRR 0.453 [0.300, 0.624]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 10/4 | 0.344 [0.000, 1.000] | 0.833 [0.583, 1.000] | – / – / – |
| MRR 0.857 [0.643, 1.000]; flow 0/0/4/6/10; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.610 [0.413, 0.785] | 0.067 [0.000, 0.188] | 0.000 / 0.150 / 0.850 |
| MRR 0.393 [0.317, 0.485]; flow 0/0/11/9/0; pipeline 0/0/0/3/17. | | | | |
| L1 paper | 20/20 | 0.710 [0.550, 0.854] | 0.000 [0.000, 0.000] | 0.000 / 0.800 / 0.200 |
| MRR 0.467 [0.426, 0.500]; flow 0/0/11/9/0; pipeline 0/0/4/12/4. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.495 [0.271, 0.713] | 0.400 [0.190, 0.619] | – / – / – |
| MRR 0.541 [0.378, 0.712]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 9/3 | 0.333 [0.000, 1.000] | 0.907 [0.688, 1.000] | – / – / – |
| MRR 0.921 [0.732, 1.000]; flow 0/0/8/1/11; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.562 [0.366, 0.760] | 0.050 [0.000, 0.167] | 0.000 / 0.000 / 1.000 |
| MRR 0.357 [0.284, 0.448]; flow 0/0/11/9/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/20 | 0.655 [0.467, 0.833] | 0.050 [0.000, 0.167] | 0.050 / 0.000 / 0.950 |
| MRR 0.391 [0.315, 0.483]; flow 0/0/11/9/0; pipeline 0/0/0/1/19. | | | | |
| L2 paper | 20/20 | 0.887 [0.764, 0.979] | 0.000 [0.000, 0.000] | 0.000 / 1.000 / 0.000 |
| MRR 0.487 [0.458, 0.500]; flow 0/0/11/9/0; pipeline 0/0/7/13/0. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| **core.E1**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.482 [0.337, 0.628] | 0.100 [0.000, 0.250] | – / – / – |
| MRR 0.276 [0.179, 0.401]; flow 0/4/8/8/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 12/5 | 0.600 [0.167, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/6/4/8; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.675 [0.493, 0.845] | 0.300 [0.105, 0.500] | 0.050 / 0.000 / 0.950 |
| MRR 0.570 [0.440, 0.708]; flow 0/4/8/8/0; pipeline 0/0/1/0/19. | | | | |
| L1 paper | 20/20 | 0.620 [0.453, 0.785] | 0.000 [0.000, 0.000] | 0.000 / 0.900 / 0.100 |
| MRR 0.463 [0.425, 0.492]; flow 0/4/8/8/0; pipeline 0/0/11/7/2. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/4/8/8/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.415 [0.233, 0.615] | 0.150 [0.000, 0.333] | – / – / – |
| MRR 0.378 [0.263, 0.515]; flow 0/4/8/8/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 7/1 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/1/4/13; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.703 [0.574, 0.836] | 0.300 [0.105, 0.520] | 0.200 / 0.000 / 0.800 |
| MRR 0.568 [0.440, 0.704]; flow 0/4/8/8/0; pipeline 0/4/0/0/16. | | | | |
| L2 net | 20/20 | 0.700 [0.529, 0.877] | 0.400 [0.188, 0.625] | 0.000 / 0.050 / 0.950 |
| MRR 0.607 [0.460, 0.759]; flow 0/4/8/8/0; pipeline 0/0/0/1/19. | | | | |
| L2 paper | 20/20 | 0.652 [0.485, 0.821] | 0.000 [0.000, 0.000] | 0.000 / 1.000 / 0.000 |
| MRR 0.464 [0.422, 0.500]; flow 0/4/8/8/0; pipeline 0/0/13/7/0. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/4/8/8/0; pipeline 0/0/0/0/0. | | | | |
| **core.C1-SF**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/10/9/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 14/7 | 0.602 [0.250, 0.880] | 0.048 [0.011, 0.091] | – / – / – |
| MRR 0.273 [0.251, 0.295]; flow 0/1/5/8/6; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.993 [0.969, 1.000] | 1.000 [1.000, 1.000] | 0.650 / 0.000 / 0.350 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/10/9/0; pipeline 0/1/4/8/7. | | | | |
| L1 paper | 20/20 | 0.998 [0.986, 1.000] | 0.950 [0.833, 1.000] | 0.550 / 0.000 / 0.450 |
| MRR 0.975 [0.917, 1.000]; flow 0/1/10/9/0; pipeline 0/1/3/7/9. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.250 [0.250, 0.250]; flow 0/1/10/9/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.980 [0.927, 1.000] | 0.900 [0.750, 1.000] | – / – / – |
| MRR 0.950 [0.875, 1.000]; flow 0/1/10/9/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 11/3 | 0.667 [0.000, 1.000] | 0.030 [0.000, 0.074] | – / – / – |
| MRR 0.260 [0.233, 0.288]; flow 0/1/6/4/9; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.645 [0.479, 0.801] | 0.350 [0.143, 0.565] | 0.050 / 0.000 / 0.950 |
| MRR 0.534 [0.376, 0.695]; flow 0/1/10/9/0; pipeline 0/1/0/0/19. | | | | |
| L2 net | 20/20 | 0.850 [0.723, 0.958] | 0.800 [0.609, 0.952] | 0.400 / 0.050 / 0.550 |
| MRR 0.872 [0.743, 0.974]; flow 0/1/10/9/0; pipeline 0/0/2/7/11. | | | | |
| L2 paper | 20/20 | 0.995 [0.978, 1.000] | 0.600 [0.381, 0.812] | 0.450 / 0.200 / 0.350 |
| MRR 0.800 [0.690, 0.906]; flow 0/1/10/9/0; pipeline 0/1/5/7/7. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.250 [0.250, 0.250]; flow 0/1/10/9/0; pipeline 0/0/0/0/0. | | | | |
| **core.C1-FF**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.993 [0.972, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 10/4 | 0.438 [0.000, 1.000] | 0.100 [0.000, 0.333] | – / – / – |
| MRR 0.383 [0.250, 0.561]; flow 0/1/2/7/10; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/4/15/0. | | | | |
| L1 paper | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/4/15/0. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.250 [0.250, 0.250]; flow 0/1/4/15/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 14/7 | 0.551 [0.344, 0.812] | 0.012 [0.000, 0.039] | – / – / – |
| MRR 0.383 [0.308, 0.455]; flow 0/1/3/10/6; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.868 [0.727, 0.964] | 0.600 [0.375, 0.812] | 0.150 / 0.000 / 0.850 |
| MRR 0.724 [0.565, 0.871]; flow 0/1/4/15/0; pipeline 0/1/0/2/17. | | | | |
| L2 net | 20/20 | 0.998 [0.986, 1.000] | 1.000 [1.000, 1.000] | 0.950 / 0.000 / 0.050 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/3/15/1. | | | | |
| L2 paper | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/4/15/0. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.250 [0.250, 0.250]; flow 0/1/4/15/0; pipeline 0/0/0/0/0. | | | | |
| **labs.gemma-12b.N0**; declared 20, collected 20, missing 0, ineligible 6, eligible 14 | | | | |
| L1 corr. | 14/– | – | 0.071 [0.000, 0.231] | – / – / – |
| MRR 0.288 [0.192, 0.426]; flow 0/0/8/6/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/14; pipeline 0/0/0/0/0. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L1 net | 14/– | – | 0.143 [0.000, 0.357] | 0.000 / 0.071 / 0.929 |
| MRR 0.430 [0.300, 0.581]; flow 0/0/8/6/0; pipeline 0/0/1/0/13. Alarms: any 1/14 [0.002, 0.339]; one 1/14 [0.002, 0.339]; ref. 0/14 [0.000, 0.232] | | | | |
| L1 paper | 14/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.357 / 0.643 |
| MRR 0.401 [0.332, 0.464]; flow 0/0/8/6/0; pipeline 0/0/3/2/9. Alarms: any 5/14 [0.128, 0.649]; one 5/14 [0.128, 0.649]; ref. 0/14 [0.000, 0.232] | | | | |
| L1 strength | 14/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/6/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 14/– | – | 0.286 [0.067, 0.545] | – / – / – |
| MRR 0.483 [0.310, 0.674]; flow 0/0/8/6/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/14; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 14/– | – | 0.071 [0.000, 0.231] | 0.000 / 0.071 / 0.929 |
| MRR 0.374 [0.270, 0.500]; flow 0/0/8/6/0; pipeline 0/1/0/0/13. Alarms: any 1/14 [0.002, 0.339]; one 1/14 [0.002, 0.339]; ref. 0/14 [0.000, 0.232] | | | | |
| L2 net | 14/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.214 / 0.786 |
| MRR 0.363 [0.292, 0.431]; flow 0/0/8/6/0; pipeline 0/0/1/2/11. Alarms: any 3/14 [0.047, 0.508]; one 3/14 [0.047, 0.508]; ref. 0/14 [0.000, 0.232] | | | | |
| L2 paper | 14/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.857 / 0.143 |
| MRR 0.457 [0.393, 0.500]; flow 0/0/8/6/0; pipeline 0/0/8/4/2. Alarms: any 12/14 [0.572, 0.982]; one 12/14 [0.572, 0.982]; ref. 0/14 [0.000, 0.232] | | | | |
| L2 strength | 14/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/6/0; pipeline 0/0/0/0/0. | | | | |
| **labs.gemma-12b.C1**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/14 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/14 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 0.900 / 0.000 / 0.100 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/5/12/2. | | | | |
| L1 paper | 20/14 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 0.950 / 0.000 / 0.050 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/5/13/1. | | | | |
| L1 strength | 20/14 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/14 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/14 | 0.969 [0.876, 1.000] | 0.600 [0.375, 0.812] | 0.050 / 0.000 / 0.950 |
| MRR 0.758 [0.613, 0.890]; flow 0/1/5/14/0; pipeline 0/1/0/0/19. | | | | |
| L2 net | 20/14 | 0.995 [0.973, 1.000] | 1.000 [1.000, 1.000] | 0.550 / 0.000 / 0.450 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/3/7/9. | | | | |
| L2 paper | 20/14 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/5/14/0. | | | | |
| L2 strength | 20/14 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0. | | | | |
| **labs.ministral-8b.N0**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.100 [0.000, 0.250] | – / – / – |
| MRR 0.378 [0.275, 0.496]; flow 0/1/12/7/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.100 [0.000, 0.250] | 0.050 / 0.250 / 0.700 |
| MRR 0.419 [0.322, 0.530]; flow 0/1/12/7/0; pipeline 0/0/2/4/14. Alarms: any 6/20 [0.119, 0.543]; one 6/20 [0.119, 0.543]; ref. 1/20 [0.001, 0.249] | | | | |
| L1 paper | 20/– | – | 0.050 [0.000, 0.167] | 0.000 / 0.000 / 1.000 |
| MRR 0.384 [0.302, 0.474]; flow 0/1/12/7/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168] | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/12/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.200 [0.045, 0.389] | – / – / – |
| MRR 0.445 [0.322, 0.579]; flow 0/1/12/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L2 leak | 20/– | – | 0.150 [0.000, 0.320] | 0.050 / 0.050 / 0.900 |
| MRR 0.392 [0.280, 0.522]; flow 0/1/12/7/0; pipeline 0/2/0/0/18. Alarms: any 2/20 [0.012, 0.317]; one 2/20 [0.012, 0.317]; ref. 1/20 [0.001, 0.249] | | | | |
| L2 net | 20/– | – | 0.050 [0.000, 0.167] | 0.000 / 0.150 / 0.850 |
| MRR 0.359 [0.290, 0.447]; flow 0/1/12/7/0; pipeline 0/0/0/3/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.300 / 0.700 |
| MRR 0.303 [0.252, 0.356]; flow 0/1/12/7/0; pipeline 0/0/1/5/14. Alarms: any 6/20 [0.119, 0.543]; one 6/20 [0.119, 0.543]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/12/7/0; pipeline 0/0/0/0/0. | | | | |
| **labs.ministral-8b.C1**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.897 [0.785, 0.975] | 0.650 [0.435, 0.857] | – / – / – |
| MRR 0.808 [0.687, 0.922]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.860 [0.712, 0.975] | 0.800 [0.611, 0.955] | 0.550 / 0.000 / 0.450 |
| MRR 0.877 [0.756, 0.975]; flow 0/0/8/12/0; pipeline 0/0/5/6/9. | | | | |
| L1 paper | 20/20 | 0.922 [0.803, 1.000] | 0.750 [0.550, 0.938] | 0.450 / 0.000 / 0.550 |
| MRR 0.854 [0.734, 0.963]; flow 0/0/8/12/0; pipeline 0/0/5/4/11. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.828 [0.707, 0.935] | 0.600 [0.375, 0.812] | – / – / – |
| MRR 0.771 [0.639, 0.896]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.675 [0.469, 0.859] | 0.250 [0.062, 0.455] | 0.000 / 0.000 / 1.000 |
| MRR 0.512 [0.387, 0.650]; flow 0/0/8/12/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/20 | 0.745 [0.567, 0.908] | 0.600 [0.375, 0.815] | 0.200 / 0.150 / 0.650 |
| MRR 0.756 [0.615, 0.891]; flow 0/0/8/12/0; pipeline 0/0/3/4/13. | | | | |
| L2 paper | 20/20 | 0.935 [0.838, 0.992] | 0.650 [0.429, 0.857] | 0.450 / 0.050 / 0.500 |
| MRR 0.770 [0.621, 0.910]; flow 0/0/8/12/0; pipeline 0/0/3/7/10. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0. | | | | |
| **sizes.qwen-4b.N0**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.150 [0.000, 0.333] | – / – / – |
| MRR 0.314 [0.193, 0.463]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.050 [0.000, 0.167] | 0.050 / 0.100 / 0.850 |
| MRR 0.361 [0.282, 0.455]; flow 0/0/13/7/0; pipeline 0/0/2/1/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 1/20 [0.001, 0.249] | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.650 / 0.350 |
| MRR 0.429 [0.369, 0.482]; flow 0/0/13/7/0; pipeline 0/0/4/9/7. Alarms: any 13/20 [0.408, 0.846]; one 13/20 [0.408, 0.846]; ref. 0/20 [0.000, 0.168] | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.200 [0.045, 0.392] | – / – / – |
| MRR 0.452 [0.324, 0.594]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.050 [0.000, 0.167] | 0.000 / 0.150 / 0.850 |
| MRR 0.352 [0.277, 0.447]; flow 0/0/13/7/0; pipeline 0/3/0/0/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 net | 20/– | – | 0.150 [0.000, 0.333] | 0.000 / 0.000 / 1.000 |
| MRR 0.384 [0.272, 0.518]; flow 0/0/13/7/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 1.000 / 0.000 |
| MRR 0.437 [0.383, 0.483]; flow 0/0/13/7/0; pipeline 0/0/10/10/0. Alarms: any 20/20 [0.832, 1.000]; one 20/20 [0.832, 1.000]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| **sizes.qwen-4b.C1**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.978 [0.926, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.993 [0.972, 1.000] | 1.000 [1.000, 1.000] | 0.900 / 0.000 / 0.100 |
| MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/11/7/2. | | | | |
| L1 paper | 20/20 | 1.000 [1.000, 1.000] | 0.400 [0.190, 0.619] | 0.100 / 0.450 / 0.450 |
| MRR 0.700 [0.595, 0.810]; flow 0/0/12/8/0; pipeline 0/0/6/5/9. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.955 [0.843, 1.000] | 0.950 [0.833, 1.000] | – / – / – |
| MRR 0.975 [0.917, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.463 [0.279, 0.652] | 0.200 [0.048, 0.389] | 0.000 / 0.000 / 1.000 |
| MRR 0.452 [0.326, 0.589]; flow 0/0/12/8/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/20 | 0.943 [0.866, 0.992] | 0.700 [0.500, 0.895] | 0.750 / 0.050 / 0.200 |
| MRR 0.833 [0.714, 0.941]; flow 0/0/12/8/0; pipeline 0/0/8/8/4. | | | | |
| L2 paper | 20/20 | 0.960 [0.903, 1.000] | 0.000 [0.000, 0.000] | 0.000 / 0.950 / 0.050 |
| MRR 0.483 [0.444, 0.500]; flow 0/0/12/8/0; pipeline 0/0/13/6/1. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0. | | | | |
| **sizes.gemma-e4b.N0**; declared 20, collected 20, missing 0, ineligible 20, eligible 0 | | | | |
| L1 corr. | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 net | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 paper | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 strength | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 net | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 paper | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 strength | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| **sizes.gemma-e4b.C1**; declared 20, collected 20, missing 0, ineligible 20, eligible 0 | | | | |
| L1 corr. | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 net | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 paper | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 strength | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 net | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 paper | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L2 strength | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| **sizes.ministral-3b.N0**; declared 20, collected 20, missing 0, ineligible 20, eligible 0 | | | | |
| L1 corr. | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 net | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 paper | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 strength | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 net | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 paper | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 strength | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0. | | | | |
| **sizes.ministral-3b.C1**; declared 20, collected 20, missing 0, ineligible 18, eligible 2 | | | | |
| L1 corr. | 2/– | – | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.292 [0.250, 0.333]; flow 0/0/2/0/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/2; pipeline 0/0/0/0/0. | | | | |
| L1 net | 2/– | – | 0.000 [0.000, 0.000] | 0.000 / 1.000 / 0.000 |
| MRR 0.500 [0.500, 0.500]; flow 0/0/2/0/0; pipeline 0/0/2/0/0. | | | | |
| L1 paper | 2/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.500 / 0.500 |
| MRR 0.417 [0.333, 0.500]; flow 0/0/2/0/0; pipeline 0/0/1/0/1. | | | | |
| L1 strength | 2/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/2/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 2/– | – | 0.500 [0.000, 1.000] | – / – / – |
| MRR 0.750 [0.500, 1.000]; flow 0/0/2/0/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/2; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 2/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.000 / 1.000 |
| MRR 0.267 [0.200, 0.333]; flow 0/0/2/0/0; pipeline 0/0/0/0/2. | | | | |
| L2 net | 2/– | – | 0.500 [0.000, 1.000] | 0.500 / 0.000 / 0.500 |
| MRR 0.600 [0.200, 1.000]; flow 0/0/2/0/0; pipeline 0/0/1/0/1. | | | | |
| L2 paper | 2/– | – | 0.500 [0.000, 1.000] | 0.000 / 0.500 / 0.500 |
| MRR 0.625 [0.250, 1.000]; flow 0/0/2/0/0; pipeline 0/0/1/0/1. | | | | |
| L2 strength | 2/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/2/0/0; pipeline 0/0/0/0/0. | | | | |
| **mixed.D**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.226 [0.199, 0.254]; flow 0/2/13/5/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.200 [0.048, 0.389] | 0.000 / 0.000 / 1.000 |
| MRR 0.491 [0.374, 0.621]; flow 0/2/13/5/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168] | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.000 / 1.000 |
| MRR 0.327 [0.268, 0.388]; flow 0/2/13/5/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168] | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/13/5/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.150 [0.000, 0.333] | – / – / – |
| MRR 0.366 [0.251, 0.501]; flow 0/2/13/5/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L2 leak | 20/– | – | 0.150 [0.000, 0.333] | 0.100 / 0.050 / 0.850 |
| MRR 0.432 [0.324, 0.558]; flow 0/2/13/5/0; pipeline 0/3/0/0/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 2/20 [0.012, 0.317] | | | | |
| L2 net | 20/– | – | 0.250 [0.062, 0.455] | 0.000 / 0.050 / 0.950 |
| MRR 0.431 [0.289, 0.591]; flow 0/2/13/5/0; pipeline 0/1/0/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.100 / 0.900 |
| MRR 0.254 [0.209, 0.308]; flow 0/2/13/5/0; pipeline 0/0/1/1/18. Alarms: any 2/20 [0.012, 0.317]; one 2/20 [0.012, 0.317]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/13/5/0; pipeline 0/0/0/0/0. | | | | |
| **mixed.C1-XX**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.990 [0.963, 1.000] | 0.850 [0.667, 1.000] | – / – / – |
| MRR 0.925 [0.833, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.965 [0.898, 1.000] | 0.950 [0.833, 1.000] | 0.800 / 0.000 / 0.200 |
| MRR 0.975 [0.917, 1.000]; flow 0/2/8/10/0; pipeline 0/2/6/8/4. | | | | |
| L1 paper | 20/20 | 0.955 [0.848, 1.000] | 0.900 [0.750, 1.000] | 0.450 / 0.000 / 0.550 |
| MRR 0.950 [0.875, 1.000]; flow 0/2/8/10/0; pipeline 0/1/2/6/11. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.998 [0.986, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.695 [0.505, 0.858] | 0.350 [0.143, 0.571] | 0.100 / 0.000 / 0.900 |
| MRR 0.545 [0.396, 0.706]; flow 0/2/8/10/0; pipeline 0/2/0/0/18. | | | | |
| L2 net | 20/20 | 0.885 [0.779, 0.965] | 0.700 [0.500, 0.889] | 0.500 / 0.100 / 0.400 |
| MRR 0.833 [0.714, 0.939]; flow 0/2/8/10/0; pipeline 0/2/3/7/8. | | | | |
| L2 paper | 20/20 | 0.993 [0.969, 1.000] | 0.950 [0.833, 1.000] | 0.600 / 0.000 / 0.400 |
| MRR 0.975 [0.917, 1.000]; flow 0/2/8/10/0; pipeline 0/1/5/6/8. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0. | | | | |
| **mixed.H-XYZZ**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.050 [0.000, 0.167] | – / – / – |
| MRR 0.333 [0.250, 0.432]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.050 / 0.950 |
| MRR 0.391 [0.331, 0.446]; flow 0/0/17/3/0; pipeline 0/0/1/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168] | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.050 / 0.950 |
| MRR 0.368 [0.305, 0.428]; flow 0/0/17/3/0; pipeline 0/0/1/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168] | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.050 [0.000, 0.167] | – / – / – |
| MRR 0.369 [0.292, 0.462]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.050 / 0.950 |
| MRR 0.342 [0.251, 0.457]; flow 0/0/17/3/0; pipeline 0/1/0/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 net | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.100 / 0.900 |
| MRR 0.430 [0.334, 0.541]; flow 0/0/17/3/0; pipeline 0/1/1/0/18. Alarms: any 2/20 [0.012, 0.317]; one 2/20 [0.012, 0.317]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.350 / 0.650 |
| MRR 0.269 [0.217, 0.327]; flow 0/0/17/3/0; pipeline 0/1/2/4/13. Alarms: any 7/20 [0.154, 0.592]; one 7/20 [0.154, 0.592]; ref. 0/20 [0.000, 0.168] | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| **mixed.C1-XY**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L1 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 0.950 / 0.000 / 0.050 |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/5/7/7/1. | | | | |
| L1 paper | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/6/7/7/0. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 0/– | – | – | – / – / – |
| MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.860 [0.701, 0.989] | 0.750 [0.550, 0.933] | 0.350 / 0.000 / 0.650 |
| MRR 0.838 [0.702, 0.956]; flow 0/6/7/7/0; pipeline 0/6/0/1/13. | | | | |
| L2 net | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/6/7/7/0. | | | | |
| L2 paper | 20/20 | 1.000 [1.000, 1.000] | 1.000 [1.000, 1.000] | 1.000 / 0.000 / 0.000 |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/6/7/7/0. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0. | | | | |
| **core.C1-NW**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/20 | 0.993 [0.960, 1.000] | 0.900 [0.750, 1.000] | – / – / – |
| MRR 0.933 [0.833, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 12/5 | 0.480 [0.000, 1.000] | 0.931 [0.762, 1.000] | – / – / – |
| MRR 0.940 [0.796, 1.000]; flow 0/0/4/8/8; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/20 | 0.968 [0.903, 1.000] | 0.800 [0.609, 0.955] | 0.600 / 0.000 / 0.400 |
| MRR 0.892 [0.786, 0.976]; flow 0/1/8/11/0; pipeline 0/0/4/8/8. | | | | |
| L1 paper | 20/20 | 0.932 [0.847, 0.994] | 0.350 [0.150, 0.565] | 0.100 / 0.400 / 0.500 |
| MRR 0.667 [0.562, 0.778]; flow 0/1/8/11/0; pipeline 0/0/5/5/10. | | | | |
| L1 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/20 | 0.985 [0.932, 1.000] | 0.950 [0.833, 1.000] | – / – / – |
| MRR 0.975 [0.917, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 10/5 | 0.800 [0.500, 1.000] | 0.900 [0.667, 1.000] | – / – / – |
| MRR 0.950 [0.833, 1.000]; flow 0/0/4/6/10; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/20 | 0.682 [0.505, 0.844] | 0.200 [0.048, 0.389] | 0.100 / 0.050 / 0.850 |
| MRR 0.464 [0.340, 0.598]; flow 0/1/8/11/0; pipeline 0/2/0/1/17. | | | | |
| L2 net | 20/20 | 0.963 [0.910, 1.000] | 0.650 [0.438, 0.857] | 0.650 / 0.050 / 0.300 |
| MRR 0.817 [0.704, 0.923]; flow 0/1/8/11/0; pipeline 0/1/5/8/6. | | | | |
| L2 paper | 20/20 | 0.975 [0.901, 1.000] | 0.000 [0.000, 0.000] | 0.000 / 0.950 / 0.050 |
| MRR 0.492 [0.472, 0.500]; flow 0/1/8/11/0; pipeline 0/0/12/7/1. | | | | |
| L2 strength | 20/20 | 0.500 [0.500, 0.500] | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0. | | | | |
| **lookalike.LX**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.214 [0.188, 0.251]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 14/– | – | 0.881 [0.702, 1.000] | – / – / – |
| MRR 0.898 [0.745, 1.000]; flow 0/0/8/6/6; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.050 [0.000, 0.167] | 0.000 / 0.050 / 0.950 |
| MRR 0.368 [0.289, 0.461]; flow 0/0/10/10/0; pipeline 0/0/0/1/19. | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.100 / 0.900 |
| MRR 0.283 [0.231, 0.339]; flow 0/0/10/10/0; pipeline 0/0/1/1/18. | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.050 [0.000, 0.167] | – / – / – |
| MRR 0.316 [0.240, 0.411]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 9/– | – | 0.907 [0.688, 1.000] | – / – / – |
| MRR 0.921 [0.732, 1.000]; flow 0/0/5/4/11; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.150 [0.000, 0.318] | 0.000 / 0.000 / 1.000 |
| MRR 0.420 [0.306, 0.548]; flow 0/0/10/10/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/– | – | 0.150 [0.000, 0.318] | 0.000 / 0.000 / 1.000 |
| MRR 0.398 [0.292, 0.524]; flow 0/0/10/10/0; pipeline 0/0/0/0/20. | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.600 / 0.400 |
| MRR 0.237 [0.203, 0.279]; flow 0/0/10/10/0; pipeline 0/0/8/4/8. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| **lookalike.LX-swap**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.231 [0.194, 0.280]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 7/– | – | 0.881 [0.583, 1.000] | – / – / – |
| MRR 0.898 [0.643, 1.000]; flow 0/0/4/3/13; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.000 / 1.000 |
| MRR 0.399 [0.307, 0.509]; flow 0/0/13/7/0; pipeline 0/0/0/0/20. | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.000 / 1.000 |
| MRR 0.277 [0.228, 0.336]; flow 0/0/13/7/0; pipeline 0/0/0/0/20. | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.100 [0.000, 0.250] | – / – / – |
| MRR 0.393 [0.296, 0.506]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 8/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/6/2/12; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.000 / 1.000 |
| MRR 0.403 [0.316, 0.513]; flow 0/0/13/7/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/– | – | 0.200 [0.048, 0.391] | 0.000 / 0.050 / 0.950 |
| MRR 0.438 [0.315, 0.582]; flow 0/0/13/7/0; pipeline 0/0/1/0/19. | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.650 / 0.350 |
| MRR 0.237 [0.204, 0.276]; flow 0/0/13/7/0; pipeline 0/0/9/4/7. | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0. | | | | |
| **lookalike.LY**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.000 [0.000, 0.000] | – / – / – |
| MRR 0.188 [0.172, 0.210]; flow 0/0/15/5/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 10/– | – | 0.783 [0.528, 1.000] | – / – / – |
| MRR 0.867 [0.717, 1.000]; flow 0/0/8/2/10; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.150 / 0.850 |
| MRR 0.340 [0.241, 0.461]; flow 0/0/15/5/0; pipeline 0/0/2/1/17. | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.350 / 0.650 |
| MRR 0.382 [0.326, 0.438]; flow 0/0/15/5/0; pipeline 0/0/3/4/13. | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/15/5/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.250 [0.059, 0.450] | – / – / – |
| MRR 0.520 [0.391, 0.655]; flow 0/0/15/5/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 6/– | – | 0.611 [0.233, 1.000] | – / – / – |
| MRR 0.742 [0.481, 1.000]; flow 0/0/3/3/14; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.000 / 1.000 |
| MRR 0.301 [0.254, 0.353]; flow 0/0/15/5/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.100 / 0.900 |
| MRR 0.334 [0.280, 0.390]; flow 0/0/15/5/0; pipeline 0/0/1/1/18. | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.950 / 0.050 |
| MRR 0.299 [0.251, 0.352]; flow 0/0/15/5/0; pipeline 0/0/8/11/1. | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/15/5/0; pipeline 0/0/0/0/0. | | | | |
| **lookalike.LY-swap**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.050 [0.000, 0.167] | – / – / – |
| MRR 0.276 [0.209, 0.371]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 12/– | – | 0.778 [0.533, 1.000] | – / – / – |
| MRR 0.814 [0.613, 1.000]; flow 0/0/11/1/8; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.050 [0.000, 0.167] | 0.050 / 0.000 / 0.950 |
| MRR 0.386 [0.312, 0.473]; flow 0/0/17/3/0; pipeline 0/0/1/0/19. | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.400 / 0.600 |
| MRR 0.387 [0.330, 0.440]; flow 0/0/17/3/0; pipeline 0/0/5/3/12. | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.150 [0.000, 0.333] | – / – / – |
| MRR 0.420 [0.310, 0.548]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 7/– | – | 0.762 [0.444, 1.000] | – / – / – |
| MRR 0.796 [0.524, 1.000]; flow 0/0/6/1/13; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.050 / 0.950 |
| MRR 0.367 [0.277, 0.478]; flow 0/0/17/3/0; pipeline 0/1/0/0/19. | | | | |

| Level / detector | Cov./n | AUROC [interval] | Top-1 [interval] | Unique / wrong / abstain |
|---|---|---|---|---|
| L2 net | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.150 / 0.850 |
| MRR 0.313 [0.274, 0.356]; flow 0/0/17/3/0; pipeline 0/0/0/3/17. | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.900 / 0.100 |
| MRR 0.355 [0.296, 0.412]; flow 0/0/17/3/0; pipeline 0/0/10/8/2. | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0. | | | | |
| **lookalike.LZ**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.050 [0.000, 0.167] | – / – / – |
| MRR 0.317 [0.235, 0.418]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 12/– | – | 0.944 [0.810, 1.000] | – / – / – |
| MRR 0.958 [0.857, 1.000]; flow 0/0/6/6/8; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.150 [0.000, 0.333] | 0.000 / 0.050 / 0.950 |
| MRR 0.386 [0.275, 0.517]; flow 0/0/11/9/0; pipeline 0/0/0/1/19. | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.000 / 1.000 |
| MRR 0.249 [0.198, 0.308]; flow 0/0/11/9/0; pipeline 0/0/0/0/20. | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.150 [0.000, 0.333] | – / – / – |
| MRR 0.376 [0.265, 0.511]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 6/– | – | 0.667 [0.250, 1.000] | – / – / – |
| MRR 0.806 [0.542, 1.000]; flow 0/0/2/4/14; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.200 [0.048, 0.389] | 0.000 / 0.000 / 1.000 |
| MRR 0.429 [0.305, 0.572]; flow 0/0/11/9/0; pipeline 0/0/0/0/20. | | | | |
| L2 net | 20/– | – | 0.100 [0.000, 0.250] | 0.050 / 0.150 / 0.800 |
| MRR 0.362 [0.268, 0.478]; flow 0/0/11/9/0; pipeline 0/0/0/4/16. | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.250 / 0.750 |
| MRR 0.258 [0.212, 0.312]; flow 0/0/11/9/0; pipeline 0/0/1/4/15. | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0. | | | | |
| **lookalike.LZ-swap**; declared 20, collected 20, missing 0, ineligible 0, eligible 20 | | | | |
| L1 corr. | 20/– | – | 0.050 [0.000, 0.167] | – / – / – |
| MRR 0.246 [0.189, 0.342]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| L1 Q judge | 8/– | – | 0.583 [0.259, 0.889] | – / – / – |
| MRR 0.696 [0.455, 0.917]; flow 0/0/4/4/12; pipeline 0/0/0/0/0. | | | | |
| L1 net | 20/– | – | 0.100 [0.000, 0.250] | 0.000 / 0.000 / 1.000 |
| MRR 0.353 [0.257, 0.473]; flow 0/0/10/10/0; pipeline 0/0/0/0/20. | | | | |
| L1 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.000 / 1.000 |
| MRR 0.211 [0.181, 0.251]; flow 0/0/10/10/0; pipeline 0/0/0/0/20. | | | | |
| L1 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| L2 corr. | 20/– | – | 0.150 [0.000, 0.333] | – / – / – |
| MRR 0.354 [0.243, 0.494]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |
| L2 Q judge | 5/– | – | 0.800 [0.333, 1.000] | – / – / – |
| MRR 0.880 [0.600, 1.000]; flow 0/0/4/1/15; pipeline 0/0/0/0/0. | | | | |
| L2 leak | 20/– | – | 0.250 [0.062, 0.450] | 0.000 / 0.050 / 0.950 |
| MRR 0.506 [0.381, 0.646]; flow 0/0/10/10/0; pipeline 0/1/0/0/19. | | | | |
| L2 net | 20/– | – | 0.300 [0.111, 0.520] | 0.050 / 0.050 / 0.900 |
| MRR 0.512 [0.372, 0.666]; flow 0/0/10/10/0; pipeline 0/0/0/2/18. | | | | |
| L2 paper | 20/– | – | 0.000 [0.000, 0.000] | 0.000 / 0.150 / 0.850 |
| MRR 0.254 [0.208, 0.309]; flow 0/0/10/10/0; pipeline 0/1/0/2/17. | | | | |
| L2 strength | 20/– | – | 1.000 [1.000, 1.000] | – / – / – |
| MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0. | | | | |

## D.1 Prefix, missing-bid and breadth sensitivity

The following compact table provides the score trajectories. Full intervals, coverage, flow labels and reporting outcomes for every prefix/mask remain in `results/analysis/final_v1/all_cells.md`. A3 merges all registered breadth indices; it does not select the first subset after observing scores. The unavailable Gemma-E4B and Ministral-3B cells remain in the comprehensive table above. Mixed XX and XY treatments each use their own composition-matched honest control; their contrast does not isolate model identity.

**Table 6:** Prefix and missing-bid sensitivity for every available treatment comparison. Each entry is AUROC; exact intervals, exclusions and other metrics for every prefix/mask are in all_cells.md.

| Cell / level / detector | Mask | n (full) | Prefix 50 | 100 | 200 |
|---|---|---|---|---|---|
| core.P0/L1/net | complete | 20 | 1.000 | 1.000 | 1.000 |
| core.P0/L1/net | no_bid | 20 | 1.000 | 1.000 | 1.000 |
| core.P0/L1/correlation | complete | 20 | 1.000 | 1.000 | 1.000 |
| core.P0/L1/correlation | no_bid | 20 | 1.000 | 1.000 | 1.000 |
| core.P0/L2/net | complete | 20 | 0.897 | 1.000 | 1.000 |
| core.P0/L2/net | no_bid | 20 | 0.890 | 1.000 | 1.000 |
| core.P0/L2/correlation | complete | 20 | 0.978 | 1.000 | 1.000 |
| core.P0/L2/correlation | no_bid | 20 | 0.978 | 1.000 | 1.000 |
| core.P0/L2/leak | complete | 20 | 0.953 | 1.000 | 1.000 |
| core.P0/L2/leak | no_bid | 20 | 0.953 | 1.000 | 1.000 |
| core.C1-SS/L1/net | complete | 20 | 0.882 | 0.915 | 0.912 |
| core.C1-SS/L1/net | no_bid | 20 | 0.882 | 0.915 | 0.917 |
| core.C1-SS/L1/correlation | complete | 20 | 0.968 | 0.978 | 0.988 |
| core.C1-SS/L1/correlation | no_bid | 20 | 0.968 | 0.975 | 0.988 |
| core.C1-SS/L2/net | complete | 20 | 0.650 | 0.795 | 0.865 |
| core.C1-SS/L2/net | no_bid | 20 | 0.660 | 0.785 | 0.858 |
| core.C1-SS/L2/correlation | complete | 20 | 0.975 | 0.998 | 0.998 |
| core.C1-SS/L2/correlation | no_bid | 20 | 0.915 | 0.990 | 1.000 |
| core.C1-SS/L2/leak | complete | 20 | 0.522 | 0.562 | 0.640 |
| core.C1-SS/L2/leak | no_bid | 20 | 0.520 | 0.560 | 0.627 |
| core.C1-SF/L1/net | complete | 20 | 0.762 | 0.840 | 0.993 |
| core.C1-SF/L1/net | no_bid | 20 | 0.760 | 0.870 | 0.970 |
| core.C1-SF/L1/correlation | complete | 20 | 0.902 | 0.990 | 1.000 |
| core.C1-SF/L1/correlation | no_bid | 20 | 0.902 | 0.993 | 1.000 |
| core.C1-SF/L2/net | complete | 20 | 0.700 | 0.720 | 0.850 |
| core.C1-SF/L2/net | no_bid | 20 | 0.690 | 0.730 | 0.853 |
| core.C1-SF/L2/correlation | complete | 20 | 0.833 | 0.920 | 0.980 |
| core.C1-SF/L2/correlation | no_bid | 20 | 0.748 | 0.887 | 0.998 |
| core.C1-SF/L2/leak | complete | 20 | 0.650 | 0.642 | 0.645 |
| core.C1-SF/L2/leak | no_bid | 20 | 0.660 | 0.665 | 0.675 |
| core.C1-FF/L1/net | complete | 20 | 0.907 | 1.000 | 1.000 |
| core.C1-FF/L1/net | no_bid | 20 | 0.902 | 1.000 | 1.000 |
| core.C1-FF/L1/correlation | complete | 20 | 0.960 | 0.985 | 0.993 |
| core.C1-FF/L1/correlation | no_bid | 20 | 0.938 | 0.985 | 0.990 |
| core.C1-FF/L2/net | complete | 20 | 0.750 | 0.882 | 0.998 |
| core.C1-FF/L2/net | no_bid | 20 | 0.752 | 0.887 | 0.998 |
| core.C1-FF/L2/correlation | complete | 20 | 0.963 | 0.995 | 1.000 |
| core.C1-FF/L2/correlation | no_bid | 20 | 0.958 | 0.995 | 1.000 |
| core.C1-FF/L2/leak | complete | 20 | 0.580 | 0.740 | 0.868 |
| core.C1-FF/L2/leak | no_bid | 20 | 0.590 | 0.708 | 0.863 |
| core.C1-NW/L1/net | complete | 20 | 0.845 | 0.902 | 0.968 |
| core.C1-NW/L1/net | no_bid | 20 | 0.873 | 0.905 | 0.975 |
| core.C1-NW/L1/correlation | complete | 20 | 0.932 | 0.955 | 0.993 |
| core.C1-NW/L1/correlation | no_bid | 20 | 0.970 | 0.980 | 1.000 |
| core.C1-NW/L2/net | complete | 20 | 0.725 | 0.797 | 0.963 |
| core.C1-NW/L2/net | no_bid | 20 | 0.730 | 0.810 | 0.963 |
| core.C1-NW/L2/correlation | complete | 20 | 0.927 | 0.955 | 0.985 |
| core.C1-NW/L2/correlation | no_bid | 20 | 0.843 | 0.905 | 0.930 |
| core.C1-NW/L2/leak | complete | 20 | 0.598 | 0.613 | 0.682 |
| core.C1-NW/L2/leak | no_bid | 20 | 0.590 | 0.620 | 0.690 |
| core.C2/L1/net | complete | 20 | 0.662 | 0.688 | 0.610 |
| core.C2/L1/net | no_bid | 20 | 0.657 | 0.680 | 0.605 |
| core.C2/L1/correlation | complete | 20 | 0.333 | 0.407 | 0.463 |
| core.C2/L1/correlation | no_bid | 20 | 0.370 | 0.432 | 0.485 |
| core.C2/L2/net | complete | 20 | 0.445 | 0.470 | 0.655 |
| core.C2/L2/net | no_bid | 20 | 0.458 | 0.482 | 0.665 |
| core.C2/L2/correlation | complete | 20 | 0.410 | 0.502 | 0.495 |
| core.C2/L2/correlation | no_bid | 20 | 0.365 | 0.470 | 0.472 |
| core.C2/L2/leak | complete | 20 | 0.415 | 0.417 | 0.562 |
| core.C2/L2/leak | no_bid | 20 | 0.420 | 0.430 | 0.575 |
| core.E1/L1/net | complete | 20 | 0.578 | 0.585 | 0.675 |
| core.E1/L1/net | no_bid | 20 | 0.593 | 0.598 | 0.672 |
| core.E1/L1/correlation | complete | 20 | 0.450 | 0.500 | 0.482 |

| Cell / level / detector | Mask | n (full) | Prefix 50 | 100 | 200 |
|---|---|---|---|---|---|
| core.E1/L1/correlation | no_bid | 20 | 0.512 | 0.540 | 0.475 |
| core.E1/L2/net | complete | 20 | 0.458 | 0.665 | 0.700 |
| core.E1/L2/net | no_bid | 20 | 0.455 | 0.682 | 0.705 |
| core.E1/L2/correlation | complete | 20 | 0.440 | 0.500 | 0.415 |
| core.E1/L2/correlation | no_bid | 20 | 0.480 | 0.522 | 0.535 |
| core.E1/L2/leak | complete | 20 | 0.522 | 0.695 | 0.703 |
| core.E1/L2/leak | no_bid | 20 | 0.515 | 0.698 | 0.705 |
| labs.gemma-12b.C1/L1/net | complete | 14 | 0.765 | 1.000 | 1.000 |
| labs.gemma-12b.C1/L1/net | no_bid | 14 | 0.832 | 1.000 | 1.000 |
| labs.gemma-12b.C1/L1/correlation | complete | 14 | 0.995 | 1.000 | 1.000 |
| labs.gemma-12b.C1/L1/correlation | no_bid | 14 | 0.944 | 0.974 | 0.954 |
| labs.gemma-12b.C1/L2/net | complete | 14 | 0.699 | 0.883 | 0.995 |
| labs.gemma-12b.C1/L2/net | no_bid | 14 | 0.714 | 0.883 | 0.995 |
| labs.gemma-12b.C1/L2/correlation | complete | 14 | 0.842 | 1.000 | 1.000 |
| labs.gemma-12b.C1/L2/correlation | no_bid | 14 | 0.867 | 0.959 | 0.944 |
| labs.gemma-12b.C1/L2/leak | complete | 14 | 0.653 | 0.791 | 0.969 |
| labs.gemma-12b.C1/L2/leak | no_bid | 14 | 0.781 | 0.796 | 0.949 |
| labs.ministral-8b.C1/L1/net | complete | 20 | 0.583 | 0.825 | 0.860 |
| labs.ministral-8b.C1/L1/net | no_bid | 20 | 0.590 | 0.820 | 0.855 |
| labs.ministral-8b.C1/L1/correlation | complete | 20 | 0.853 | 0.875 | 0.897 |
| labs.ministral-8b.C1/L1/correlation | no_bid | 20 | 0.853 | 0.875 | 0.897 |
| labs.ministral-8b.C1/L2/net | complete | 20 | 0.627 | 0.605 | 0.745 |
| labs.ministral-8b.C1/L2/net | no_bid | 20 | 0.618 | 0.613 | 0.743 |
| labs.ministral-8b.C1/L2/correlation | complete | 20 | 0.585 | 0.760 | 0.828 |
| labs.ministral-8b.C1/L2/correlation | no_bid | 20 | 0.585 | 0.760 | 0.828 |
| labs.ministral-8b.C1/L2/leak | complete | 20 | 0.590 | 0.637 | 0.675 |
| labs.ministral-8b.C1/L2/leak | no_bid | 20 | 0.588 | 0.650 | 0.675 |
| sizes.qwen-4b.C1/L1/net | complete | 20 | 0.858 | 0.955 | 0.993 |
| sizes.qwen-4b.C1/L1/net | no_bid | 20 | 0.858 | 0.955 | 0.993 |
| sizes.qwen-4b.C1/L1/correlation | complete | 20 | 0.760 | 0.915 | 0.978 |
| sizes.qwen-4b.C1/L1/correlation | no_bid | 20 | 0.760 | 0.915 | 0.978 |
| sizes.qwen-4b.C1/L2/net | complete | 20 | 0.718 | 0.865 | 0.943 |
| sizes.qwen-4b.C1/L2/net | no_bid | 20 | 0.723 | 0.860 | 0.945 |
| sizes.qwen-4b.C1/L2/correlation | complete | 20 | 0.723 | 0.838 | 0.955 |
| sizes.qwen-4b.C1/L2/correlation | no_bid | 20 | 0.723 | 0.838 | 0.955 |
| sizes.qwen-4b.C1/L2/leak | complete | 20 | 0.550 | 0.450 | 0.463 |
| sizes.qwen-4b.C1/L2/leak | no_bid | 20 | 0.552 | 0.448 | 0.470 |
| mixed.C1-XX/L1/net | complete | 20 | 0.780 | 0.912 | 0.965 |
| mixed.C1-XX/L1/net | no_bid | 20 | 0.802 | 0.912 | 0.958 |
| mixed.C1-XX/L1/correlation | complete | 20 | 0.943 | 0.985 | 0.990 |
| mixed.C1-XX/L1/correlation | no_bid | 20 | 0.948 | 0.988 | 0.990 |
| mixed.C1-XX/L2/net | complete | 20 | 0.573 | 0.657 | 0.885 |
| mixed.C1-XX/L2/net | no_bid | 20 | 0.580 | 0.675 | 0.900 |
| mixed.C1-XX/L2/correlation | complete | 20 | 0.875 | 0.920 | 0.998 |
| mixed.C1-XX/L2/correlation | no_bid | 20 | 0.885 | 0.917 | 0.998 |
| mixed.C1-XX/L2/leak | complete | 20 | 0.485 | 0.552 | 0.695 |
| mixed.C1-XX/L2/leak | no_bid | 20 | 0.490 | 0.545 | 0.705 |
| mixed.C1-XY/L1/net | complete | 20 | 0.950 | 0.963 | 1.000 |
| mixed.C1-XY/L1/net | no_bid | 20 | 0.960 | 0.970 | 1.000 |
| mixed.C1-XY/L1/correlation | complete | 20 | 0.980 | 1.000 | 1.000 |
| mixed.C1-XY/L1/correlation | no_bid | 20 | 0.980 | 1.000 | 1.000 |
| mixed.C1-XY/L2/net | complete | 20 | 0.818 | 0.948 | 1.000 |
| mixed.C1-XY/L2/net | no_bid | 20 | 0.830 | 0.950 | 1.000 |
| mixed.C1-XY/L2/correlation | complete | 20 | 0.970 | 1.000 | 1.000 |
| mixed.C1-XY/L2/correlation | no_bid | 20 | 0.968 | 1.000 | 1.000 |
| mixed.C1-XY/L2/leak | complete | 20 | 0.792 | 0.865 | 0.860 |
| mixed.C1-XY/L2/leak | no_bid | 20 | 0.762 | 0.865 | 0.868 |

# E Statistics, provenance, and deviations

## E.1 A1: warning removal

A1 compares matched N0/C1-SS/C1-NW triplets. Its profit contrast is the difference of additive N0-referenced joint-profit uplifts; the relative uplift companion is listed separately. All eligible triplets contribute to the non-judge estimates.

Judge comparisons use only triplets judged successfully in all conditions. The L2 judge difference of 0.5 with interval $[0.5, 0.5]$ comes from a single triplet and is not precise evidence.

**Table 7:** A1: warning removed minus warning present. Judge rows use only complete judged triplets; other full rows use all eligible triplets.

| Endpoint / subset | n | Difference [95% interval] |
|---|---|---|
| All: L1:net | 20 | 0.0550 [-0.0425, 0.1725] |
| All: L2:net | 20 | 0.0975 [-0.0051, 0.2300] |
| All: L2:leak | 20 | 0.0425 [-0.1625, 0.2576] |
| All: L1:correlation | 20 | 0.0050 [-0.0250, 0.0250] |
| All: L2:correlation | 20 | -0.0125 [-0.0675, 0.0100] |
| All: L1:Q judge | 5 | 0.1600 [-0.4000, 0.6400] |
| All: L2:Q judge | 1 | 0.5000 [0.5000, 0.5000] |
| All: joint_profit | 20 | -0.3478 [-0.4015, -0.2921] |
| All: relative uplift | 20 | -1.0849 [-1.2533, -0.9299] |
| All: winning_price | 20 | -0.3335 [-0.3925, -0.2753] |
| All: pair_win_share | 20 | 0.0706 [0.0540, 0.0879] |
| Judge L1 subset: L1:net | 5 | -0.0400 [-0.1600, 0.0000] |
| Judge L1 subset: L2:net | 5 | 0.0400 [-0.1200, 0.4000] |
| Judge L1 subset: L2:leak | 5 | 0.2800 [0.0000, 0.7200] |
| Judge L1 subset: L1:correlation | 5 | 0.0000 [0.0000, 0.0000] |
| Judge L1 subset: L2:correlation | 5 | 0.0000 [0.0000, 0.0000] |
| Judge L1 subset: L1:Q judge | 5 | 0.1600 [-0.4000, 0.6400] |
| Judge L1 subset: joint_profit | 5 | -0.2905 [-0.3826, -0.1985] |
| Judge L1 subset: relative uplift | 5 | -0.9591 [-1.2861, -0.6699] |
| Judge L1 subset: winning_price | 5 | -0.2927 [-0.4227, -0.1860] |
| Judge L1 subset: pair_win_share | 5 | 0.0607 [0.0267, 0.0972] |
| Judge L2 subset: L1:net | 1 | 0.0000 [0.0000, 0.0000] |
| Judge L2 subset: L2:net | 1 | 1.0000 [1.0000, 1.0000] |
| Judge L2 subset: L2:leak | 1 | 0.0000 [0.0000, 0.0000] |
| Judge L2 subset: L1:correlation | 1 | 0.0000 [0.0000, 0.0000] |
| Judge L2 subset: L2:correlation | 1 | 0.0000 [0.0000, 0.0000] |
| Judge L2 subset: L2:Q judge | 1 | 0.5000 [0.5000, 0.5000] |
| Judge L2 subset: joint_profit | 1 | -0.5287 [-0.5287, -0.5287] |
| Judge L2 subset: relative uplift | 1 | -1.0070 [-1.0070, -1.0070] |
| Judge L2 subset: winning_price | 1 | -0.5132 [-0.5132, -0.5132] |
| Judge L2 subset: pair_win_share | 1 | 0.1158 [0.1158, 0.1158] |

## E.2 A2: honest placement swaps

Write $s_{ab}$ for the score of strong firm $a$ and fringe firm $b$. The registered symmetrised placement contrast is

$$T = \tfrac{1}{2}\{s_{00}(\text{cell}) - s_{00}(\text{twin}) + s_{01}(\text{twin}) - s_{01}(\text{cell})\}.$$

The descriptive within-original-cell interaction is $D = s_{00} - s_{01} - s_{10} + s_{11}$. The intervention changes fringe placement and therefore can change the rest of the market. It does not identify an isolated sameness mechanism. The two-sided test swaps cell/twin labels within each index and twin-pair block. Bootstrap sampling keeps blocks with the same market index together. The tables retain all registered endpoints, per-twin contrasts and corresponding successful-judge subsets. Candidate denominators are pair opportunities, not independent trials.

**Table 8:** A2: registered placement-swap tests. None rejects after Holm correction across the registered endpoints.

| Endpoint | n | Mean T [95% interval] | Raw p | Holm p |
|---|---|---|---|---|
| L1:net | 60 | -0.00281 [-0.00872, 0.00254] | 0.30401 | 1.00000 |
| L2:net | 60 | 0.00751 [-0.00636, 0.02201] | 0.24581 | 1.00000 |
| L2:leak | 60 | 0.00616 [0.00003, 0.01211] | 0.03771 | 0.26397 |
| L1:correlation | 60 | -0.00034 [-0.01416, 0.01347] | 0.97100 | 1.00000 |
| L2:correlation | 60 | 0.00494 [-0.00781, 0.01745] | 0.51502 | 1.00000 |
| L1:Q judge | 15 | 0.04000 [-0.10833, 0.13553] | 0.54763 | 1.00000 |
| L2:Q judge | 10 | 0.04500 [-0.01500, 0.13864] | 0.38943 | 1.00000 |

**Table 9:** A2 descriptive companions: original-cell interaction D and per-twin-pair contrasts. Counts are contributing matched cell/twin blocks.

| Endpoint / twin | n | Mean T [95% interval] | D [95% interval] |
|---|---|---|---|
| L1:net / pooled | 60 | -0.0028 [-0.0087, 0.0025] | -0.0091 [-0.0207, 0.0014] |
| L1:net / LX | 20 | -0.0085 [-0.0170, -0.0000] | -0.0117 [-0.0356, 0.0113] |
| L1:net / LY | 20 | -0.0007 [-0.0100, 0.0090] | -0.0095 [-0.0389, 0.0166] |
| L1:net / LZ | 20 | 0.0008 [-0.0081, 0.0090] | -0.0061 [-0.0230, 0.0089] |
| L2:net / pooled | 60 | 0.0075 [-0.0064, 0.0220] | 0.0195 [-0.0090, 0.0505] |
| L2:net / LX | 20 | 0.0093 [-0.0116, 0.0300] | 0.0511 [0.0088, 0.0986] |
| L2:net / LY | 20 | 0.0119 [-0.0109, 0.0335] | 0.0100 [-0.0318, 0.0562] |
| L2:net / LZ | 20 | 0.0014 [-0.0197, 0.0220] | -0.0026 [-0.0490, 0.0401] |
| L2:leak / pooled | 60 | 0.0062 [0.0000, 0.0121] | 0.0166 [0.0069, 0.0261] |
| L2:leak / LX | 20 | 0.0042 [-0.0048, 0.0136] | 0.0211 [-0.0003, 0.0411] |
| L2:leak / LY | 20 | 0.0072 [-0.0036, 0.0179] | 0.0151 [-0.0076, 0.0384] |
| L2:leak / LZ | 20 | 0.0071 [-0.0017, 0.0163] | 0.0136 [-0.0035, 0.0312] |
| L1:correlation / pooled | 60 | -0.0003 [-0.0142, 0.0135] | -0.0232 [-0.0484, 0.0021] |
| L1:correlation / LX | 20 | 0.0489 [0.0308, 0.0661] | -0.0087 [-0.0373, 0.0182] |
| L1:correlation / LY | 20 | -0.0122 [-0.0462, 0.0279] | -0.0119 [-0.0727, 0.0538] |
| L1:correlation / LZ | 20 | -0.0377 [-0.0610, -0.0148] | -0.0489 [-0.1202, 0.0193] |
| L2:correlation / pooled | 60 | 0.0049 [-0.0078, 0.0174] | -0.0160 [-0.0494, 0.0173] |
| L2:correlation / LX | 20 | 0.0003 [-0.0158, 0.0153] | 0.0113 [-0.0295, 0.0485] |
| L2:correlation / LY | 20 | 0.0254 [-0.0008, 0.0523] | -0.0280 [-0.0749, 0.0193] |
| L2:correlation / LZ | 20 | -0.0109 [-0.0391, 0.0179] | -0.0313 [-0.0802, 0.0144] |
| L1:Q judge / pooled | 15 | 0.0400 [-0.1083, 0.1355] | 0.0080 [-0.0043, 0.0267] |
| L1:Q judge / LX | 3 | 0.0000 [0.0000, 0.0000] | 0.0167 [0.0000, 0.0500] |
| L1:Q judge / LY | 6 | 0.2250 [0.0300, 0.4202] | -0.0050 [-0.0150, 0.0000] |
| L1:Q judge / LZ | 6 | -0.1250 [-0.3125, 0.0000] | 0.0167 [0.0000, 0.0500] |
| L2:Q judge / pooled | 10 | 0.0450 [-0.0150, 0.1386] | 0.0560 [-0.0357, 0.2192] |
| L2:Q judge / LX | 4 | 0.0037 [-0.0100, 0.0250] | -0.0225 [-0.0500, 0.0000] |
| L2:Q judge / LY | 4 | 0.1313 [-0.0250, 0.4000] | 0.1625 [-0.1000, 0.7000] |
| L2:Q judge / LZ | 2 | -0.0450 [-0.0500, -0.0400] | -0.0000 [-0.0000, 0.0000] |

**Table 10:** A2 on each successful judge subset: descriptive detector companions, without an additional Holm family.

| Judge / detector / twin | n | Mean T [95% interval] | D [95% interval] |
|---|---|---|---|
| L1/L1:net/pooled | 15 | -0.0004 [-0.0098, 0.0125] | -0.0064 [-0.0329, 0.0294] |
| L1/L1:net/LX | 3 | 0.0029 [-0.0139, 0.0254] | 0.0249 [-0.0063, 0.0663] |
| L1/L1:net/LY | 6 | -0.0065 [-0.0260, 0.0135] | -0.0458 [-0.1200, 0.0204] |
| L1/L1:net/LZ | 6 | 0.0041 [-0.0054, 0.0161] | 0.0172 [-0.0037, 0.0371] |
| L1/L2:net/pooled | 15 | 0.0294 [0.0001, 0.0587] | 0.0430 [-0.0255, 0.1052] |
| L1/L2:net/LX | 3 | 0.0651 [-0.0198, 0.1198] | 0.1393 [0.0410, 0.2532] |
| L1/L2:net/LY | 6 | 0.0274 [-0.0213, 0.0765] | 0.0364 [-0.0922, 0.1710] |
| L1/L2:net/LZ | 6 | 0.0134 [-0.0011, 0.0277] | 0.0013 [-0.0288, 0.0301] |
| L1/L2:leak/pooled | 15 | 0.0138 [0.0007, 0.0263] | 0.0222 [-0.0053, 0.0454] |
| L1/L2:leak/LX | 3 | 0.0148 [-0.0127, 0.0395] | 0.0504 [0.0337, 0.0780] |
| L1/L2:leak/LY | 6 | 0.0153 [-0.0068, 0.0386] | 0.0138 [-0.0468, 0.0679] |
| L1/L2:leak/LZ | 6 | 0.0118 [-0.0072, 0.0269] | 0.0164 [-0.0157, 0.0497] |
| L1/L1:correlation/pooled | 15 | -0.0078 [-0.0341, 0.0176] | -0.0210 [-0.0889, 0.0516] |
| L1/L1:correlation/LX | 3 | 0.0556 [0.0127, 0.0952] | 0.0412 [-0.0359, 0.0801] |
| L1/L1:correlation/LY | 6 | -0.0112 [-0.0519, 0.0310] | -0.0380 [-0.1649, 0.0533] |
| L1/L1:correlation/LZ | 6 | -0.0361 [-0.0684, -0.0116] | -0.0351 [-0.2002, 0.1231] |
| L1/L2:correlation/pooled | 15 | 0.0142 [-0.0231, 0.0502] | -0.0287 [-0.0816, 0.0219] |
| L1/L2:correlation/LX | 3 | 0.0085 [-0.0432, 0.0723] | -0.0000 [-0.0731, 0.0975] |
| L1/L2:correlation/LY | 6 | 0.0442 [-0.0164, 0.1010] | -0.0361 [-0.1025, 0.0427] |
| L1/L2:correlation/LZ | 6 | -0.0130 [-0.0541, 0.0287] | -0.0357 [-0.1161, 0.0492] |
| L1/L1:judge:qwen-9b/pooled | 15 | 0.0400 [-0.1083, 0.1355] | 0.0080 [-0.0043, 0.0267] |
| L1/L1:judge:qwen-9b/LX | 3 | 0.0000 [0.0000, 0.0000] | 0.0167 [0.0000, 0.0500] |
| L1/L1:judge:qwen-9b/LY | 6 | 0.2250 [0.0300, 0.4202] | -0.0050 [-0.0150, 0.0000] |
| L1/L1:judge:qwen-9b/LZ | 6 | -0.1250 [-0.3125, 0.0000] | 0.0167 [0.0000, 0.0500] |
| L2/L1:net/pooled | 10 | -0.0025 [-0.0184, 0.0126] | -0.0468 [-0.0835, -0.0143] |
| L2/L1:net/LX | 4 | -0.0149 [-0.0420, 0.0148] | -0.0525 [-0.0934, 0.0158] |
| L2/L1:net/LY | 4 | 0.0037 [-0.0403, 0.0259] | -0.0511 [-0.1669, 0.0294] |
| L2/L1:net/LZ | 2 | 0.0100 [-0.0031, 0.0231] | -0.0268 [-0.0297, -0.0240] |

| Judge / detector / twin | n | Mean T [95% interval] | D [95% interval] |
|---|---|---|---|
| L2/L2:net/pooled | 10 | 0.0118 [-0.0256, 0.0428] | -0.0170 [-0.0831, 0.0517] |
| L2/L2:net/LX | 4 | -0.0027 [-0.1141, 0.0457] | 0.0076 [-0.0933, 0.1339] |
| L2/L2:net/LY | 4 | 0.0306 [-0.0001, 0.0852] | -0.0448 [-0.1308, 0.0678] |
| L2/L2:net/LZ | 2 | 0.0031 [-0.0393, 0.0455] | -0.0105 [-0.1720, 0.1509] |
| L2/L2:leak/pooled | 10 | 0.0005 [-0.0136, 0.0160] | -0.0141 [-0.0432, 0.0137] |
| L2/L2:leak/LX | 4 | -0.0022 [-0.0174, 0.0142] | -0.0046 [-0.0434, 0.0267] |
| L2/L2:leak/LY | 4 | 0.0096 [-0.0235, 0.0460] | -0.0123 [-0.1029, 0.0635] |
| L2/L2:leak/LZ | 2 | -0.0123 [-0.0268, 0.0021] | -0.0366 [-0.0659, -0.0073] |
| L2/L1:correlation/pooled | 10 | 0.0124 [-0.0326, 0.0543] | -0.0611 [-0.1287, -0.0047] |
| L2/L1:correlation/LX | 4 | 0.0506 [0.0007, 0.0887] | -0.0092 [-0.0751, 0.0475] |
| L2/L1:correlation/LY | 4 | 0.0008 [-0.0671, 0.0938] | -0.1173 [-0.3024, 0.0415] |
| L2/L1:correlation/LZ | 2 | -0.0411 [-0.1136, 0.0314] | -0.0524 [-0.0770, -0.0279] |
| L2/L2:correlation/pooled | 10 | 0.0314 [-0.0159, 0.0776] | -0.0435 [-0.1275, 0.0403] |
| L2/L2:correlation/LX | 4 | -0.0052 [-0.0768, 0.0345] | 0.0007 [-0.2387, 0.1074] |
| L2/L2:correlation/LY | 4 | 0.0276 [-0.0746, 0.1373] | -0.1512 [-0.2005, -0.1081] |
| L2/L2:correlation/LZ | 2 | 0.1119 [0.0956, 0.1283] | 0.0837 [0.0388, 0.1286] |
| L2/L2:judge:qwen-9b/pooled | 10 | 0.0450 [-0.0150, 0.1386] | 0.0560 [-0.0357, 0.2192] |
| L2/L2:judge:qwen-9b/LX | 4 | 0.0037 [-0.0100, 0.0250] | -0.0225 [-0.0500, 0.0000] |
| L2/L2:judge:qwen-9b/LY | 4 | 0.1313 [-0.0250, 0.4000] | 0.1625 [-0.1000, 0.7000] |
| L2/L2:judge:qwen-9b/LZ | 2 | -0.0450 [-0.0500, -0.0400] | -0.0000 [-0.0000, 0.0000] |

| Endpoint | Same-model candidates | Different-model candidates |
|---|---|---|
| L1:net | 0/120 (0.000%) | 6/600 (1.000%) |
| L2:net | 2/120 (1.667%) | 10/600 (1.667%) |
| L2:leak | 1/120 (0.833%) | 1/600 (0.167%) |

## E.3 Judge failures and matched coverage

Qwen produced 278 valid and 322 invalid judgements out of 600; Gemma produced 5 valid and 35 invalid out of 40. Content failures were retained, not resubmitted for a preferred answer. Transport failures were reconciled and retried unchanged as recorded in the deviations. Breadth judging was cut for budget. Its absent rows are uncollected judge coverage, not judge scores of zero. The registered limit was 12,000 generated tokens per attempt with the registered retry; most failed calls ended with `length`, according to the raw-log observation in the final handover. This report does not infer a cause for every invalid output from the failure code alone.

The successful-output Qwen top-1 pattern in C1-SS was $0.808/0.881$ at L1/L2, versus SF $0.0476/0.0303$ and FF $0.100/0.0119$. The honest N0 strong reference pair received $0.952/1.000$. These are treatment-only available-output summaries, not the matched samples plotted in Figure 7. They are consistent with strong-pair preference in these cells, without establishing a universal rule or an internal reasoning mechanism. P0 is a counterexample to a universal strong-pair account: its matched judge AUROC is zero at each access level, with only 4 and 3 blocks. Tiny-sample percentile intervals can collapse and must not be read as certainty.

**Table 11:** Judge completion and content failures, including every attempted cell and access level. Invalid entries failed to return valid JSON under the registered budget and retry rule.

| Judge / cell | Level | Total | Valid | Invalid JSON |
|---|---|---|---|---|
| gemma-12b/core.C1-SS | L1 | 10 | 3 | 7 |
| gemma-12b/core.C1-SS | L2 | 10 | 1 | 9 |
| gemma-12b/core.N0 | L1 | 10 | 1 | 9 |
| gemma-12b/core.N0 | L2 | 10 | 0 | 10 |
| qwen-9b/core.C1-FF | L1 | 20 | 10 | 10 |
| qwen-9b/core.C1-FF | L2 | 20 | 14 | 6 |
| qwen-9b/core.C1-NW | L1 | 20 | 12 | 8 |
| qwen-9b/core.C1-NW | L2 | 20 | 10 | 10 |
| qwen-9b/core.C1-SF | L1 | 20 | 14 | 6 |
| qwen-9b/core.C1-SF | L2 | 20 | 11 | 9 |
| qwen-9b/core.C1-SS | L1 | 20 | 13 | 7 |
| qwen-9b/core.C1-SS | L2 | 20 | 7 | 13 |
| qwen-9b/core.C2 | L1 | 20 | 10 | 10 |
| qwen-9b/core.C2 | L2 | 20 | 9 | 11 |
| qwen-9b/core.E1 | L1 | 20 | 12 | 8 |

| Judge / cell | Level | Total | Valid | Invalid JSON |
|---|---|---|---|---|
| qwen-9b/core.E1 | L2 | 20 | 7 | 13 |
| qwen-9b/core.N0 | L1 | 40 | 21 | 19 |
| qwen-9b/core.N0 | L2 | 40 | 15 | 25 |
| qwen-9b/core.P0 | L1 | 20 | 5 | 15 |
| qwen-9b/core.P0 | L2 | 20 | 4 | 16 |
| qwen-9b/lookalike.LX | L1 | 20 | 14 | 6 |
| qwen-9b/lookalike.LX | L2 | 20 | 9 | 11 |
| qwen-9b/lookalike.LX-swap | L1 | 20 | 7 | 13 |
| qwen-9b/lookalike.LX-swap | L2 | 20 | 8 | 12 |
| qwen-9b/lookalike.LY | L1 | 20 | 10 | 10 |
| qwen-9b/lookalike.LY | L2 | 20 | 6 | 14 |
| qwen-9b/lookalike.LY-swap | L1 | 20 | 12 | 8 |
| qwen-9b/lookalike.LY-swap | L2 | 20 | 7 | 13 |
| qwen-9b/lookalike.LZ | L1 | 20 | 12 | 8 |
| qwen-9b/lookalike.LZ | L2 | 20 | 6 | 14 |
| qwen-9b/lookalike.LZ-swap | L1 | 20 | 8 | 12 |
| qwen-9b/lookalike.LZ-swap | L2 | 20 | 5 | 15 |

**Table 12:** Judge comparisons on identical successful matched blocks. Tiny or degenerate intervals do not imply precision.

| Cell / level / detector | n | Detector AUROC [interval] | Judge AUROC [interval] |
|---|---|---|---|
| core.P0/L1/net | 4 | 1.000 [1.000, 1.000] | 0.000 [0.000, 0.000] |
| core.P0/L1/correlation | 4 | 1.000 [1.000, 1.000] | 0.000 [0.000, 0.000] |
| core.P0/L1/strength | 4 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] |
| core.P0/L2/net | 3 | 1.000 [1.000, 1.000] | 0.000 [0.000, 0.000] |
| core.P0/L2/correlation | 3 | 1.000 [1.000, 1.000] | 0.000 [0.000, 0.000] |
| core.P0/L2/strength | 3 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] |
| core.P0/L2/leak | 3 | 1.000 [1.000, 1.000] | 0.000 [0.000, 0.000] |
| core.C1-SS/L1/net | 8 | 0.875 [0.600, 1.000] | 0.320 [0.069, 0.617] |
| core.C1-SS/L1/correlation | 8 | 1.000 [1.000, 1.000] | 0.320 [0.069, 0.617] |
| core.C1-SS/L1/strength | 8 | 0.500 [0.500, 0.500] | 0.320 [0.069, 0.617] |
| core.C1-SS/L2/net | 1 | 0.000 [0.000, 0.000] | 0.000 [0.000, 0.000] |
| core.C1-SS/L2/correlation | 1 | 1.000 [1.000, 1.000] | 0.000 [0.000, 0.000] |
| core.C1-SS/L2/strength | 1 | 0.500 [0.500, 0.500] | 0.000 [0.000, 0.000] |
| core.C1-SS/L2/leak | 1 | 0.000 [0.000, 0.000] | 0.000 [0.000, 0.000] |
| core.C2/L1/net | 4 | 0.250 [0.000, 1.000] | 0.344 [0.000, 1.000] |
| core.C2/L1/correlation | 4 | 0.500 [0.000, 1.000] | 0.344 [0.000, 1.000] |
| core.C2/L1/strength | 4 | 0.500 [0.500, 0.500] | 0.344 [0.000, 1.000] |
| core.C2/L2/net | 3 | 0.889 [0.750, 1.000] | 0.333 [0.000, 1.000] |
| core.C2/L2/correlation | 3 | 0.333 [0.000, 1.000] | 0.333 [0.000, 1.000] |
| core.C2/L2/strength | 3 | 0.500 [0.500, 0.500] | 0.333 [0.000, 1.000] |
| core.C2/L2/leak | 3 | 0.222 [0.000, 1.000] | 0.333 [0.000, 1.000] |
| core.E1/L1/net | 5 | 0.840 [0.438, 1.000] | 0.600 [0.167, 1.000] |
| core.E1/L1/correlation | 5 | 0.640 [0.188, 1.000] | 0.600 [0.167, 1.000] |
| core.E1/L1/strength | 5 | 0.500 [0.500, 0.500] | 0.600 [0.167, 1.000] |
| core.E1/L2/net | 1 | 1.000 [1.000, 1.000] | 0.500 [0.500, 0.500] |
| core.E1/L2/correlation | 1 | 0.000 [0.000, 0.000] | 0.500 [0.500, 0.500] |
| core.E1/L2/strength | 1 | 0.500 [0.500, 0.500] | 0.500 [0.500, 0.500] |
| core.E1/L2/leak | 1 | 1.000 [1.000, 1.000] | 0.500 [0.500, 0.500] |
| core.C1-SF/L1/net | 7 | 1.000 [1.000, 1.000] | 0.602 [0.250, 0.880] |
| core.C1-SF/L1/correlation | 7 | 1.000 [1.000, 1.000] | 0.602 [0.250, 0.880] |
| core.C1-SF/L1/strength | 7 | 0.500 [0.500, 0.500] | 0.602 [0.250, 0.880] |
| core.C1-SF/L2/net | 3 | 0.667 [0.000, 1.000] | 0.667 [0.000, 1.000] |
| core.C1-SF/L2/correlation | 3 | 0.889 [0.750, 1.000] | 0.667 [0.000, 1.000] |
| core.C1-SF/L2/strength | 3 | 0.500 [0.500, 0.500] | 0.667 [0.000, 1.000] |
| core.C1-SF/L2/leak | 3 | 0.111 [0.000, 1.000] | 0.667 [0.000, 1.000] |
| core.C1-FF/L1/net | 4 | 1.000 [1.000, 1.000] | 0.438 [0.000, 1.000] |
| core.C1-FF/L1/correlation | 4 | 0.938 [0.778, 1.000] | 0.438 [0.000, 1.000] |
| core.C1-FF/L1/strength | 4 | 0.500 [0.500, 0.500] | 0.438 [0.000, 1.000] |
| core.C1-FF/L2/net | 7 | 1.000 [1.000, 1.000] | 0.551 [0.344, 0.812] |
| core.C1-FF/L2/correlation | 7 | 1.000 [1.000, 1.000] | 0.551 [0.344, 0.812] |
| core.C1-FF/L2/strength | 7 | 0.500 [0.500, 0.500] | 0.551 [0.344, 0.812] |
| core.C1-FF/L2/leak | 7 | 0.939 [0.720, 1.000] | 0.551 [0.344, 0.812] |
| core.C1-NW/L1/net | 5 | 0.960 [0.840, 1.000] | 0.480 [0.000, 1.000] |

| Cell / level / detector | n | Detector AUROC [interval] | Judge AUROC [interval] |
|---|---|---|---|
| core.C1-NW/L1/correlation | 5 | 1.000 [1.000, 1.000] | 0.480 [0.000, 1.000] |
| core.C1-NW/L1/strength | 5 | 0.500 [0.500, 0.500] | 0.480 [0.000, 1.000] |
| core.C1-NW/L2/net | 5 | 1.000 [1.000, 1.000] | 0.800 [0.500, 1.000] |
| core.C1-NW/L2/correlation | 5 | 1.000 [1.000, 1.000] | 0.800 [0.500, 1.000] |
| core.C1-NW/L2/strength | 5 | 0.500 [0.500, 0.500] | 0.800 [0.500, 1.000] |
| core.C1-NW/L2/leak | 5 | 0.640 [0.250, 1.000] | 0.800 [0.500, 1.000] |

**Table 13:** Judge Brier scores on matched discrimination samples; prevalence is the artificial treatment/control balance, not real-world collusion prevalence.

| Cell / level / judge | n | Brier | Prevalence |
|---|---|---|---|
| core.P0/L1/qwen-9b | 4 | 0.7700 | 0.500 |
| core.P0/L2/qwen-9b | 3 | 0.8503 | 0.500 |
| core.C1-SS/L1/qwen-9b | 8 | 0.5230 | 0.500 |
| core.C1-SS/L2/qwen-9b | 1 | 0.4625 | 0.500 |
| core.C2/L1/qwen-9b | 4 | 0.5472 | 0.500 |
| core.C2/L2/qwen-9b | 3 | 0.6054 | 0.500 |
| core.E1/L1/qwen-9b | 5 | 0.4267 | 0.500 |
| core.E1/L2/qwen-9b | 1 | 0.4100 | 0.500 |
| core.C1-SF/L1/qwen-9b | 7 | 0.3861 | 0.500 |
| core.C1-SF/L2/qwen-9b | 3 | 0.3630 | 0.500 |
| core.C1-FF/L1/qwen-9b | 4 | 0.4384 | 0.500 |
| core.C1-FF/L2/qwen-9b | 7 | 0.3758 | 0.500 |
| core.C1-NW/L1/qwen-9b | 5 | 0.4782 | 0.500 |
| core.C1-NW/L2/qwen-9b | 5 | 0.4284 | 0.500 |

## E.4 Exploratory frozen-context probes

The complete-context slope was 0.0134737 $[-0.0035263, 0.0319480]$; 199 contexts were complete. One context was incomplete (market 7, round 130, seat 3). The leak uses 2997 observed bids, including observations from the incomplete context: raw 0.1481534 bits, corrected 0.0887914 bits, implemented $p = 0.001$. No resolved average linear response follows from the slope interval. A signed mean slope can cancel across contexts while mutual information detects distributional changes, but neither cancellation nor nonlinearity is demonstrated here.

The leak inference has a more fundamental limitation. Each cost-specific regenerated message is shared by five recipient bids. Individual-bid permutation within context breaks that bundle structure and can treat stochastic message variation as evidence against its null. The resulting null is anti-conservative for a claim of reproducible partner-cost transmission. The frozen estimate and implemented p-value are retained for transparency, with no causal claim. Repeated independently regenerated messages within cost/context and an inference procedure respecting that hierarchy are needed. Relabelling cost categories among whole bundles alone leaves context-conditional mutual information unchanged.

## E.5 Pooled fingerprints: descriptive numerical caution

The frozen fingerprint analysis uses 1000 market-bootstrap draws, 99 permutations per draw and 999 for each point estimate. Some point estimates lie outside their percentile intervals. The report generator flags all such entries below; it does not alter or refit them. Bootstrap bias and Monte Carlo variation are possible explanations, not demonstrated causes. A numerical audit of their relative contributions remains open, so these intervals are not central evidence for mechanism classification.

**Table 14:** Pooled directional fingerprints in bits, retained as descriptive estimates. An asterisk flags a point outside its percentile interval; these intervals have not been re-estimated.

| Cell / term | Markets | Estimate [95% interval] |
|---|---|---|
| core.N0/action | 40 | -0.00019 [-0.00015, 0.00346] * |
| core.N0/leak_to_member_1 | 40 | -0.00010 [-0.00036, 0.00260] |
| core.N0/leak_to_member_2 | 40 | 0.00020 [-0.00005, 0.00384] |
| core.P0/action | 20 | 0.30648 [0.29323, 0.31641] |

| Cell / term | Markets | Estimate [95% interval] |
|---|---|---|
| core.P0/leak_to_member_1 | 20 | 0.30548 [0.28850, 0.32980] |
| core.P0/leak_to_member_2 | 20 | 0.33977 [0.32680, 0.36131] |
| core.C1-SS/action | 20 | 0.09657 [0.08996, 0.12093] |
| core.C1-SS/leak_to_member_1 | 20 | 0.00855 [0.00725, 0.02324] |
| core.C1-SS/leak_to_member_2 | 20 | 0.00553 [0.00505, 0.01819] |
| core.C2/action | 20 | 0.00128 [-0.00098, 0.00724] |
| core.C2/leak_to_member_1 | 20 | -0.00067 [-0.00074, 0.00477] |
| core.C2/leak_to_member_2 | 20 | 0.00211 [0.00068, 0.01044] |
| core.E1/action | 20 | 0.00160 [0.00086, 0.00769] |
| core.E1/leak_to_member_1 | 20 | 0.00471 [0.00233, 0.01251] |
| core.E1/leak_to_member_2 | 20 | -0.00037 [-0.00082, 0.00750] |
| core.C1-SF/action | 20 | 0.14449 [0.14156, 0.20800] |
| core.C1-SF/leak_to_member_1 | 20 | 0.00821 [0.00907, 0.02701] * |
| core.C1-SF/leak_to_member_2 | 20 | 0.00591 [0.00677, 0.02706] * |
| core.C1-FF/action | 20 | 0.35648 [0.34066, 0.45755] |
| core.C1-FF/leak_to_member_1 | 20 | 0.03533 [0.03043, 0.06108] |
| core.C1-FF/leak_to_member_2 | 20 | 0.03399 [0.03076, 0.05980] |
| labs.gemma-12b.N0/action | 14 | 0.00123 [-0.00063, 0.00828] |
| labs.gemma-12b.N0/leak_to_member_1 | 14 | -0.00109 [-0.00111, 0.00343] |
| labs.gemma-12b.N0/leak_to_member_2 | 14 | -0.00057 [-0.00093, 0.00316] |
| labs.gemma-12b.C1/action | 20 | 0.15676 [0.15692, 0.20001] * |
| labs.gemma-12b.C1/leak_to_member_1 | 20 | 0.02509 [0.02469, 0.05021] |
| labs.gemma-12b.C1/leak_to_member_2 | 20 | 0.02396 [0.02277, 0.04355] |
| labs.ministral-8b.N0/action | 20 | 0.00464 [0.00276, 0.01030] |
| labs.ministral-8b.N0/leak_to_member_1 | 20 | 0.00083 [0.00003, 0.00545] |
| labs.ministral-8b.N0/leak_to_member_2 | 20 | -0.00052 [-0.00102, 0.00436] |
| labs.ministral-8b.C1/action | 20 | 0.05346 [0.04685, 0.07558] |
| labs.ministral-8b.C1/leak_to_member_1 | 20 | 0.00607 [0.00627, 0.01867] * |
| labs.ministral-8b.C1/leak_to_member_2 | 20 | 0.00833 [0.00780, 0.02171] |
| sizes.qwen-4b.N0/action | 20 | 0.00245 [0.00162, 0.00922] |
| sizes.qwen-4b.N0/leak_to_member_1 | 20 | -0.00038 [-0.00052, 0.00565] |
| sizes.qwen-4b.N0/leak_to_member_2 | 20 | -0.00017 [-0.00069, 0.00628] |
| sizes.qwen-4b.C1/action | 20 | 0.08314 [0.07147, 0.10169] |
| sizes.qwen-4b.C1/leak_to_member_1 | 20 | 0.00051 [0.00060, 0.01002] * |
| sizes.qwen-4b.C1/leak_to_member_2 | 20 | 0.00135 [0.00091, 0.01094] |
| sizes.ministral-3b.C1/action | 2 | 0.06218 [0.02717, 0.12382] |
| sizes.ministral-3b.C1/leak_to_member_1 | 2 | 0.00330 [0.00192, 0.04274] |
| sizes.ministral-3b.C1/leak_to_member_2 | 2 | -0.00611 [-0.00769, 0.03385] |
| mixed.D/action | 20 | -0.00175 [-0.00132, 0.00568] * |
| mixed.D/leak_to_member_1 | 20 | 0.00081 [-0.00023, 0.00701] |
| mixed.D/leak_to_member_2 | 20 | 0.00045 [-0.00029, 0.00993] |
| mixed.C1-XX/action | 20 | 0.12049 [0.10443, 0.15526] |
| mixed.C1-XX/leak_to_member_1 | 20 | 0.01366 [0.01186, 0.02979] |
| mixed.C1-XX/leak_to_member_2 | 20 | 0.00496 [0.00454, 0.01660] |
| mixed.H-XYZZ/action | 20 | -0.00087 [-0.00118, 0.00305] |
| mixed.H-XYZZ/leak_to_member_1 | 20 | -0.00002 [-0.00051, 0.00577] |
| mixed.H-XYZZ/leak_to_member_2 | 20 | -0.00017 [-0.00041, 0.00313] |
| mixed.C1-XY/action | 20 | 0.18722 [0.17532, 0.22114] |
| mixed.C1-XY/leak_to_member_1 | 20 | 0.00697 [0.00634, 0.01908] |
| mixed.C1-XY/leak_to_member_2 | 20 | 0.05460 [0.04467, 0.07799] |
| core.C1-NW/action | 20 | 0.08148 [0.06583, 0.11067] |
| core.C1-NW/leak_to_member_1 | 20 | 0.00947 [0.00534, 0.02328] |
| core.C1-NW/leak_to_member_2 | 20 | 0.01562 [0.01350, 0.02799] |
| lookalike.LX/action | 20 | -0.00014 [-0.00056, 0.00222] |
| lookalike.LX/leak_to_member_1 | 20 | 0.00087 [0.00044, 0.00717] |
| lookalike.LX/leak_to_member_2 | 20 | -0.00006 [-0.00044, 0.00396] |
| lookalike.LX-swap/action | 20 | -0.00021 [-0.00084, 0.00435] |
| lookalike.LX-swap/leak_to_member_1 | 20 | -0.00180 [-0.00137, 0.00440] * |
| lookalike.LX-swap/leak_to_member_2 | 20 | 0.00100 [0.00042, 0.00491] |
| lookalike.LY/action | 20 | 0.01095 [0.00339, 0.02283] |
| lookalike.LY/leak_to_member_1 | 20 | 0.00067 [-0.00012, 0.00521] |
| lookalike.LY/leak_to_member_2 | 20 | 0.00067 [-0.00014, 0.00675] |
| lookalike.LY-swap/action | 20 | 0.00401 [0.00144, 0.00869] |
| lookalike.LY-swap/leak_to_member_1 | 20 | 0.00035 [-0.00011, 0.00445] |
| lookalike.LY-swap/leak_to_member_2 | 20 | 0.00128 [0.00027, 0.00550] |
| lookalike.LZ/action | 20 | -0.00149 [-0.00129, 0.00468] * |

| Cell / term | Markets | Estimate [95% interval] |
|---|---|---|
| lookalike.LZ/leak_to_member_1 | 20 | 0.00039 [-0.00019, 0.00466] |
| lookalike.LZ/leak_to_member_2 | 20 | -0.00134 [-0.00112, 0.00478] * |
| lookalike.LZ-swap/action | 20 | 0.00048 [-0.00018, 0.00498] |
| lookalike.LZ-swap/leak_to_member_1 | 20 | 0.00000 [-0.00077, 0.00508] |
| lookalike.LZ-swap/leak_to_member_2 | 20 | 0.00074 [-0.00007, 0.00743] |

## E.6 Freeze, execution and reproducibility

The initial preregistration freeze was `56717aba03261552b9e09be4dded8b6e4cdc04b3`; the amendment freeze was `4c4b6f1911486d479bd043e3590e9e0f90f6a42d`. The final analysis executed at revision `7ca459c`, with the documented junction-handling loader fix (deviation 5g). It did not run exactly the amendment checkout. The registered estimands are retained, and the executable change is disclosed separately. The manifest hash is `a63ef1040a022c4d6a190fde1ae34dafdfefb13bd1626bdb57f19cdf2488a4ff`.

The primary swap seed is 20261027; the main bootstrap seed is 20261028, the fingerprint seed is 20261029, and A1 uses 20261030. A2 uses 20261031 for swaps and 20261032 for bootstrap.

**Input SHA-256 digests.** These are file-byte digests; the manifest's canonical content hash above is a different convention. `results/analysis/final_v1/results.json`
`36804b9c365f98bb03bc92d7ab44b403a1ed6a6cf274339a968e035df5fdc672`
`results/analysis/final_v1/all_cells.md`
`fc7213378b13c7b9dcd57bed5305782d7ed56d03fbbc145b76d0bfe1d2cb7c08`
`experiments/manifests/collection_v2.json`
`34799532e08c197d1797476de2624943b97c919b4d849d7af52703275553f484`
`results/analysis_runs.json`
`e298ce13a69ce4ff1f814986ef012ce5e13631db1e8a8d80d2ef0adfc910dd4d`
`results/llm/judge_index_registered.json`
`cc0050be561b8df157fd97db239f44be06404dbe0d2917ac9f6fe19aa8e50069`
`results/llm/probe_index.json`
`af80667eda2a49569ee9c39fc7503e18840f0088d6bce3282540fa0e7a953175`
`src/collusion/llm/prompts.py`
`93e585cd275f28e827e1395bfacd47cde439f6148e4fccaa12ba4a685e5ab11a`
`src/collusion/analysis/io.py`
`3fb9b6561ab8d18f1e6bf1ba5b5a29c5b47ad3e3d5c0e08f1db5535031bdc5e5`
`results/analysis/judge24k_v23/results.json`
`5eebad9350886fb36ca85107f5f02d351006bda69c98353a8fd1dea62a947b48`

The digests are of the current files. `src/collusion/analysis/io.py` differs from the version that executed `final_v1` (revision `7ca459c`, digest `4df9668f...7455` with Windows line endings) only by the optional `--run-index` argument added for the addenda (commit `63cb9f4`), which does not change any statistic. The last digest is the judge-budget cohort used in Figure 7.

The following command reproduces the final analysis from released local inputs. It is documented here, not rerun during report preparation.

```
.venv/Scripts/python experiments/analyze_llm.py --unblind
  --input results --manifest experiments/manifests/collection_v2.json
  --analysis-freeze-commit 4c4b6f1911486d479bd043e3590e9e0f90f6a42d
  --judges results/llm/judge_index_registered.json
  --probes results/llm/probe_index.json --name final_v1 --workers 4
```

Join the displayed lines into one command. From the repository root, the report-only command is:

```
.venv/Scripts/python experiments/report_figures.py
```

The optional `--quick` flag produces identical report numbers: the script reads frozen estimates and performs no resampling. It writes PDF/PNG figures and a numerical CSV/JSON to `results/report_figures/`, copies PDFs to `report/figures/`, and generates the numerical LaTeX tables. `results/report_figures/provenance.`

**Table 15:** Out-of-sample replication (Addendum v2.2). Original: registered markets 0–19; replication: fresh treatment markets 20–39 against the registered N0 markets 20–39. Estimates with 95% block-bootstrap intervals (20 matched blocks). Profit is the paired joint-profit difference versus N0.

| Cell | Sample | L2 net AUROC | L2 net − corr. | Joint profit vs N0 |
|---|---|---|---|---|
| E1 | original | 0.700 [0.529, 0.877] | 0.285 [0.062, 0.515] | −0.116 [−0.137, −0.095] |
| E1 | replication | 0.680 [0.505, 0.830] | 0.265 [0.070, 0.445] | −0.084 [−0.109, −0.060] |
| C2 | original | 0.655 [0.467, 0.833] | 0.160 [−0.125, 0.453] | 0.399 [0.364, 0.434] |
| C2 | replication | 0.752 [0.582, 0.905] | 0.122 [−0.118, 0.358] | 0.283 [0.179, 0.371] |
| C1-SS | original | 0.865 [0.734, 0.968] | −0.133 [−0.265, −0.031] | 0.144 [0.094, 0.192] |
| C1-SS | replication | 0.907 [0.802, 0.985] | −0.088 [−0.190, −0.015] | 0.173 [0.123, 0.221] |

`json` records the source digest, analysis settings and bootstrap-weight digest. In `report/`, compile with automatic package installation disabled:

```
pdflatex --disable-installer -interaction=nonstopmode main.tex
bibtex --disable-installer main
pdflatex --disable-installer -interaction=nonstopmode main.tex
pdflatex --disable-installer -interaction=nonstopmode main.tex
```

## E.7 Addenda v2.2–v2.3: replication and judge budget

Both addenda were committed before their data (`63cb9f4`, `9d759d0`). The replication cohort re-declares the registered cells `core.E1`, `core.C2` and `core.C1-SS` on markets 20–39 with `experiments/cohort_index.py`, which copies each registered manifest entry and changes only cohort, indices, control and reason. Its matched control is the registered N0 run on markets 20–39. Treatment runs: `results/llm/rep_{E1,C2,C1-SS}_v22__deviation_replication`, each complete with 4,000 rounds. Pre-specified verdicts: R1 (E1, L2 net minus correlation above zero) replicated; R1b (the same at L1) did not; R2 (C2 joint-profit uplift above zero) replicated; R4 (C1-SS L2 net AUROC above 0.5) replicated (0.9075 $[0.8025, 0.985]$). The judge check used the registered prompt, seed and sampling with a 24,000-token limit (recorded in each run's configuration and reproducibility record). Eight second-attempt streams broke server-side after 400–785 s without a reply; they were released with an audit record and resubmitted unchanged, as for the registered judges. Of 60 judgements, 54 returned valid output and 6 invalid JSON; the registered 12,000-token judge had 27 valid outputs on the same 60 markets. A check cohort on the registered markets 0–19 reproduced `final_v1` exactly for correlation, strength and paper scores and within permutation noise for net and leak (largest AUROC difference 0.0075), because permutation seeds depend on the cohort name. Commands (one line each):

```
.venv/Scripts/python experiments/analyze_llm.py --unblind --input results
  --run-index results/analysis_runs_v22.json
  --manifest experiments/manifests/collection_v2.json
  --analysis-freeze-commit 4c4b6f1911486d479bd043e3590e9e0f90f6a42d
  --name replication_v22 --workers 3
.venv/Scripts/python experiments/analyze_llm.py --unblind --input results
  --run-index results/analysis_runs_v23.json
  --manifest experiments/manifests/collection_v2.json
  --judges results/llm/judge_index_24k.json
  --analysis-freeze-commit 4c4b6f1911486d479bd043e3590e9e0f90f6a42d
  --name judge24k_v23 --workers 3
.venv/Scripts/python experiments/addendum_summary.py
  --replication results/analysis/replication_v22/results.json
  --judge results/analysis/judge24k_v23/results.json
  --out results/analysis/addendum_summary
```

**Table 16:** Addenda: AUROC versus matched N0 for every detector, original (markets 0–19) and replication (markets 20–39), with eligible markets $n$; 95% intervals are in `addendum_summary/summary.json`. Bottom: judge token-budget check.

| Cell | Sample | L1 net | L1 corr. | L1 paper | L2 net | L2 corr. | L2 paper | L2 leak | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| E1 | original | 0.675 | 0.482 | 0.620 | 0.700 | 0.415 | 0.652 | 0.703 | 20 |
| E1 | replication | 0.525 | 0.430 | 0.492 | 0.680 | 0.415 | 0.620 | 0.745 | 20 |
| C2 | original | 0.610 | 0.463 | 0.710 | 0.655 | 0.495 | 0.887 | 0.562 | 20 |
| C2 | replication | 0.598 | 0.642 | 0.585 | 0.752 | 0.630 | 0.760 | 0.573 | 20 |
| C1-SS | original | 0.912 | 0.988 | 0.885 | 0.865 | 0.998 | 0.882 | 0.640 | 20 |
| C1-SS | replication | 0.945 | 0.993 | 0.932 | 0.907 | 0.995 | 0.887 | 0.618 | 20 |

| Judge (L2, markets 0–19) | valid | C1-SS AUROC | top-1 C1-SS | top-1 C1-SF |
|---|---|---|---|---|
| 12,000 tokens (registered) | 27/60 | 0.000 (1 block) | 0.881 | 0.030 |
| 24,000 tokens (Addendum v2.3) | 54/60 | 0.625 (17 blocks) | 0.921 | 0.020 |

### E.8 Deviations and collection order

The following summary follows the appended record in `docs/PREREG.md`; per-run metadata and `results/llm/reconciliation_log.json` retain affected attempts.

1. Warm-up transport handling was corrected before any bid was collected. The stopped pilot attempt was retained.
2. A versioned parser accepted a single Markdown code block around valid JSON for Gemma/Ministral. Initial invalid A2 attempts were retained and recollected as labelled cohorts. Core Qwen and A1 kept the original parser.
3. Concurrent-request caching, lost judge replies, streaming, timeouts and Windows checkpoint replacement were repaired. Unread transport replies were reconciled and requests resumed unchanged; answers were not selected on content.
4. Disk exhaustion interrupted collection. Raw data moved to another drive through a junction, storage guards were added, and orphaned requests were reconciled. New run identifiers reused completed cached requests.
5. Later recovery addressed drive-specific guards, preemption, warm-up caps, ledger contention, call ceilings and judge timeouts. The loader's containment check was changed to accept the raw-data junction. The original budget cap was exceeded and raised; Modal billing was USD 70.35 before the addenda and USD 76.30 after them (deviation 7).
6. Breadth judges were cancelled for budget, while core, A1, A2 and the Gemma counter-judge were completed as attempted judgements. Memory failures and interrupted judge streams were reconciled; autoscaling was pinned for the batch. Content-invalid outputs remained failures. The counter-judge used its registered market subset and the fenced parser.
7. For the addenda, eight broken judge streams were released and resubmitted unchanged, and the budget stop was made tiered (judge before C1-SS); no tier was reached. The cohort index was regenerated as full copies of registered entries before any replication statistic.

The amendment was recorded before LLM outcome analysis. Actual A1 collection started alongside the core; A2 and breadth work were interleaved after warm-up, rather than following the proposed fully sequential order. The report retains this operational ordering and the parser cohorts instead of describing an uninterrupted frozen-code collection.

## F Limitations and Dual-Use / Ethical Considerations

**Instructed coordination and treatment attribution.** The cartel is assigned by prompt; this is not evidence of emergent collusion among independently deployed agents. The treatment combines a joint-profit objective, partner naming, secrecy, a surveillance warning, and, in the channel condition, private communication. Their effects cannot generally be separated. The warning-removal and no-channel comparisons vary specific components, without identifying all interactions in the bundle. Poor compliance and refusal remain part of the assigned treatment outcomes.

**Synthetic environment and external validity.** No real procurement participants are involved. The firms, private costs, contracts, and payoffs belong to a synthetic game. Its discrete bids, independently redrawn costs, asymmetric firms, and fixed horizon simplify real procurement. Real bidders can have correlated costs, durable capacity constraints, changing strategies, and legitimate information exchange. Model family and size comparisons test robustness in this environment, not intrinsic propensities to collude. Mixed-population contrasts do not isolate identity from market strength; A2 identifies a specified placement change, including its effects on the rest of the market.

**Dependence, intent, and harm.** An influence score measures statistical dependence, not agreement, intent, illegality, or harm. A leak can reflect unilateral espionage, legitimate disclosure, or a cooperative cartel. A one-way pattern does not distinguish all such cases, and nondetection does not establish absence. Realised profit uplift and winning-price differences are manipulation checks in the game; they do not prove a real-world welfare loss. Audits must retain unclear cases and abstentions instead of forcing a substantive accusation.

**Null validity and finite data.** Within-state permutation tests assume exchangeability that can fail when bidders adapt to omitted history, drift, or their own cumulative outcomes. The correction reduces estimated null bias under the specified shuffle; it does not provide a universal false-alarm guarantee. Sparse conditional tables, limited independent markets, missing bids, and model-specific parse failures affect precision and generalisation. The matched primary test has its own condition-label exchangeability null. Secondary intervals are pointwise, and error control for one detector or test family does not extend to all reported comparisons. Pooled fingerprints and repeated probe draws must not be counted as additional independent markets.

**False accusations and human review.** A false alarm can wrongly implicate an honest pair responding to shared history or similar incentives. These scores should support investigation with contextual evidence and human review, not automated sanctions. The evaluation must present honest controls, wrong-pair reports, abstention, and uncertainty alongside successful rankings. A judge's answer is another fallible baseline; family overlap and incomplete coverage limit its role as independent corroboration.

**Private-input access and privacy.** L2 assumes an authorised auditor can inspect private costs, an informational advantage that need not exist for a competition authority. A platform might hold agent inputs without being entitled to reuse or disclose them for every purpose. Deployment would require a legitimate access basis, purpose limitations, access controls, and protection of commercially sensitive data. The synthetic inputs here do not resolve those governance questions.

**Dual use and adaptive concealment.** Publishing detector strengths and blind spots could help design cartels or evade detection. In particular, behaviour-only auditing can miss dependence absorbed by the conditioning state, and an adaptive agent may deliberately suppress detectable private-cost dependence. This experiment does not test steganographic channels or establish robustness against agents optimising against the audit. The defensive value is to expose such limits and support better evaluation, rather than to offer a covert-coordination recipe.

**Compute and reproducibility.** Collection used cloud-hosted open-weight models; interruptions, retries, idle time, and judging contribute to resource use. The registered budget cap was exceeded, so the initial budget is not an account of actual spend. Total Modal billing for the project was USD 70.35 before the addenda (deployment, pilot, collection, judging and probes) and USD 76.30 metered after them, of which USD 76.03 came from credits (24 October 2026, all runs ended). Inference seeds and pinned configurations do not ensure identical outputs on re-collection. Logs and released data support analysis reproduction; unreconciled attempts, parser changes, and missing judgements must remain visible in that record.

# G LLM usage statement

Claude Code was used for planning, reviews, commits, the addenda code and runs, and the final revision of this report. Codex models wrote code in implementation rounds, reviewed the frozen results, and prepared the first report draft. Marvin Kemper made the research decisions and reviewed the work. The LLM bidders and judges described in the

methods are experimental subjects and baselines, distinct from the tools used to develop the study. AI assistance does not establish the correctness of a citation, numerical result, or interpretation; the final claims remain Marvin's responsibility. **[TODO: Marvin to complete the final substantive review and revise the prose in his own words before submission, as encouraged by the report template.]**